FRE7241 Algorithmic Portfolio Management

Lecture#6, Spring 2026

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Vector and Matrix Calculus

Let $\mathbf{v}$ and $\mathbf{w}$ be vectors, with $\mathbf{v} = \{v_i\}_{i=1}^{i=n}$, and let $\mathbb{1}$ be the unit vector, with $\mathbb{1} = \{1\}_{i=1}^{i=n}$.

Then the inner product of $\mathbf{v}$ and $\mathbf{w}$ can be written as $\mathbf{v}^T\mathbf{w} = \mathbf{w}^T\mathbf{v} = \sum_{i=1}^n v_i w_i$.

We can then express the sum of the elements of $\mathbf{v}$ as the inner product: $\mathbf{v}^T\mathbb{1} = \mathbb{1}^T\mathbf{v} = \sum_{i=1}^n v_i$.

And the sum of squares of $\mathbf{v}$ as the inner product: $\mathbf{v}^T\mathbf{v} = \sum_{i=1}^n v_i^2$.

Let $\mathbb{A}$ be a matrix, with $\mathbb{A} = \{A_{ij}\}_{i,j=1}^{i,j=n}$.

Then the inner product of matrix $\mathbb{A}$ with vectors $\mathbf{v}$ and $\mathbf{w}$ can be written as:

$$\mathbf{v}^T\mathbb{A}\,\mathbf{w} = \mathbf{w}^T\mathbb{A}^T\mathbf{v} = \sum_{i,j=1}^n A_{ij} v_i w_j$$

The derivative of a scalar variable with respect to a vector variable is a vector, for example:

$$\frac{d(\mathbf{v}^T\mathbb{1})}{d\mathbf{v}} = d_v[\mathbf{v}^T\mathbb{1}] = d_v[\mathbb{1}^T\mathbf{v}] = \mathbb{1}^T$$
$$d_v[\mathbf{v}^T\mathbf{w}] = d_v[\mathbf{w}^T\mathbf{v}] = \mathbf{w}^T$$
$$d_v[\mathbf{v}^T\mathbb{A}\,\mathbf{w}] = \mathbf{w}^T\mathbb{A}^T$$
$$d_v[\mathbf{v}^T\mathbb{A}\,\mathbf{v}] = \mathbf{v}^T\mathbb{A} + \mathbf{v}^T\mathbb{A}^T$$

The *Minimum Variance* Portfolio

The portfolio variance is equal to: $\mathbf{w}^T \mathbb{C}\, \mathbf{w}$, where $\mathbb{C}$ is the covariance matrix of returns.

If the portfolio weights $\mathbf{w}$ are subject to *linear* constraints: $\mathbf{w}^T \mathbb{1} = \sum_{i=1}^{n} w_i = 1$, then the weights that minimize the portfolio variance can be found by minimizing the *Lagrangian*:

$$\mathcal{L} = \mathbf{w}^T \mathbb{C}\, \mathbf{w} - \lambda\, (\mathbf{w}^T \mathbb{1} - 1)$$

Where λ is a *Lagrange multiplier*.

The derivative of a scalar variable with respect to a vector variable is a vector, for example:

$$d_w[\mathbf{w}^T \mathbb{1}] = d_w[\mathbb{1}^T \mathbf{w}] = \mathbb{1}^T$$

$$d_w[\mathbf{w}^T \mathbf{r}] = d_w[\mathbf{r}^T \mathbf{w}] = \mathbf{r}^T$$

$$d_w[\mathbf{w}^T \mathbb{C}\, \mathbf{w}] = \mathbf{w}^T \mathbb{C} + \mathbf{w}^T \mathbb{C}^T$$

Where $\mathbb{1}$ is the unit vector, and $\mathbf{w}^T \mathbb{1} = \mathbb{1}^T \mathbf{w} = \sum_{i=1}^{n} x_i$

The derivative of the *Lagrangian* $\mathcal{L}$ with respect to $\mathbf{w}$ is given by:

$$d_w \mathcal{L} = 2\mathbf{w}^T \mathbb{C} - \lambda \mathbb{1}^T$$

By setting the derivative to zero we find $\mathbf{w}$ equal to:

$$\mathbf{w} = \frac{1}{2} \lambda\, \mathbb{C}^{-1} \mathbb{1}$$

By multiplying the above from the left by $\mathbb{1}^T$, and using $\mathbf{w}^T \mathbb{1} = 1$, we find λ to be equal to:

$$\lambda = \frac{2}{\mathbb{1}^T \mathbb{C}^{-1} \mathbb{1}}$$

And finally the portfolio weights are then equal to:

$$\mathbf{w} = \frac{\mathbb{C}^{-1} \mathbb{1}}{\mathbb{1}^T \mathbb{C}^{-1} \mathbb{1}}$$

If the portfolio weights are subject to *quadratic* constraints: $\mathbf{w}^T \mathbf{w} = 1$ then the minimum variance weights are equal to the highest order *principal component* (with the smallest eigenvalue) of the covariance matrix $\mathbb{C}$.

Returns and Variance of the *Minimum Variance* Portfolio

The stock weights of the *minimum variance* portfolio under the constraint $\mathbf{w}^T \mathbb{1} = 1$ can be calculated using the inverse of the covariance matrix:

$$\mathbf{w} = \frac{\mathbb{C}^{-1} \mathbb{1}}{\mathbb{1}^T \mathbb{C}^{-1} \mathbb{1}}$$

The daily returns of the *minimum variance* portfolio are equal to:

$$\mathbf{r}_{mv} = \frac{\mathbf{r}^T \mathbb{C}^{-1} \mathbb{1}}{\mathbb{1}^T \mathbb{C}^{-1} \mathbb{1}} = \frac{\mathbf{r}^T \mathbb{C}^{-1} \mathbb{1}}{c_{11}}$$

Where $\mathbf{r}$ are the daily stock returns, and $c_{11} = \mathbb{1}^T \mathbb{C}^{-1} \mathbb{1}$.

The variance of the *minimum variance* portfolio is equal to:

$$\sigma_{mv}^2 = \mathbf{w}^T \mathbb{C} \, \mathbf{w} = \frac{\mathbb{1}^T \mathbb{C}^{-1} \mathbb{C} \, \mathbb{C}^{-1} \mathbb{1}}{(\mathbb{1}^T \mathbb{C}^{-1} \mathbb{1})^2} = \frac{1}{\mathbb{1}^T \mathbb{C}^{-1} \mathbb{1}} = \frac{1}{c_{11}}$$

The function `solve()` solves systems of linear equations, and also inverts square matrices.

The `%*%` operator performs *inner* (*scalar*) multiplication of vectors and matrices.

Inner multiplication multiplies the rows of one matrix with the columns of another matrix.

The function `drop()` removes any extra dimensions of length *one*.

```
> # Calculate daily ETF returns
> symbolv <- c("VTI", "IEF", "DBC")
> nstocks <- NROW(symbolv)
> retp <- na.omit(rutils::etfenv$returns[, symbolv])
> # Calculate covariance matrix of returns and its inverse
> covmat <- cov(retp)
> covinv <- solve(a=covmat)
> unitv <- rep(1, nstocks)
> # Calculate the minimum variance weights
> c11 <- drop(t(unitv) %*% covinv %*% unitv)
> weightmv <- drop(covinv %*% unitv/c11)
> # Calculate the daily minvar portfolio returns in two ways
> retmv <- (retp %*% weightmv)
> all.equal(retmv, (retp %*% covinv %*% unitv)/c11)
> # Calculate the minimum variance in three ways
> all.equal(var(retmv),
+   t(weightmv) %*% covmat %*% weightmv,
+   1/(t(unitv) %*% covinv %*% unitv))
```

The *Mean-Variance Efficient Portfolios*

A portfolio which has the smallest variance, given a target return, is a *mean-variance efficient portfolio*.

The efficient portfolio weights have two constraints: the sum of portfolio weights $\mathbf{w}$ is equal to 1: $\mathbf{w}^T \mathbb{1} = \sum_{i=1}^n w_i = 1$, and the mean portfolio return is equal to the target return r_t: $\mathbf{w}^T \bar{\mathbf{r}} = \sum_{i=1}^n w_i \bar{r}_i = r_t$.

Where $\bar{\mathbf{r}}$ are the mean stock returns.

The stock weights that minimize the portfolio variance under these constraints can be found by minimizing the *Lagrangian*:

$$\mathcal{L} = \mathbf{w}^T \mathbb{C}\, \mathbf{w} - \lambda_1 (\mathbf{w}^T \mathbb{1} - 1) - \lambda_2 (\mathbf{w}^T \mathbf{r} - r_t)$$

Where λ_1 and λ_2 are the *Lagrange multipliers*.

The derivative of the *Lagrangian* $\mathcal{L}$ with respect to $\mathbf{w}$ is given by:

$$d_w \mathcal{L} = 2\mathbf{w}^T \mathbb{C} - \lambda_1 \mathbb{1}^T - \lambda_2 \bar{\mathbf{r}}^T$$

By setting the derivative to zero we obtain the efficient portfolio weights $\mathbf{w}$:

$$\mathbf{w} = \frac{1}{2} (\lambda_1 \mathbb{C}^{-1} \mathbb{1} + \lambda_2 \mathbb{C}^{-1} \bar{\mathbf{r}})$$

By multiplying the above from the left first by $\mathbb{1}^T$, and then by $\bar{\mathbf{r}}^T$, we obtain a system of two equations for λ_1 and λ_2:

$$2\mathbb{1}^T \mathbf{w} = \lambda_1 \mathbb{1}^T \mathbb{C}^{-1} \mathbb{1} + \lambda_2 \mathbb{1}^T \mathbb{C}^{-1} \bar{\mathbf{r}} = 2$$

$$2\bar{\mathbf{r}}^T \mathbf{w} = \lambda_1 \bar{\mathbf{r}}^T \mathbb{C}^{-1} \mathbb{1} + \lambda_2 \bar{\mathbf{r}}^T \mathbb{C}^{-1} \bar{\mathbf{r}} = 2r_t$$

The above can be written in matrix notation as:

$$\begin{bmatrix} \mathbb{1}^T \mathbb{C}^{-1} \mathbb{1} & \mathbb{1}^T \mathbb{C}^{-1} \bar{\mathbf{r}} \\ \bar{\mathbf{r}}^T \mathbb{C}^{-1} \mathbb{1} & \bar{\mathbf{r}}^T \mathbb{C}^{-1} \bar{\mathbf{r}} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2r_t \end{bmatrix}$$

Or:

$$\begin{bmatrix} c_{11} & c_{r1} \\ c_{r1} & c_{rr} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \mathbb{F} \lambda = 2 \begin{bmatrix} 1 \\ r_t \end{bmatrix} = 2u$$

With $c_{11} = \mathbb{1}^T \mathbb{C}^{-1} \mathbb{1}$, $c_{r1} = \mathbb{1}^T \mathbb{C}^{-1} \bar{\mathbf{r}}$, $c_{rr} = \bar{\mathbf{r}}^T \mathbb{C}^{-1} \bar{\mathbf{r}}$, $\lambda = \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix}$, $u = \begin{bmatrix} 1 \\ r_t \end{bmatrix}$, and $\mathbb{F} = u^T \mathbb{C}^{-1} u = \begin{bmatrix} c_{11} & c_{r1} \\ c_{r1} & c_{rr} \end{bmatrix}$.

The *Lagrange multipliers* can be solved as:

$$\lambda = 2\mathbb{F}^{-1} u$$

The *Efficient Portfolio* Weights

The efficient portfolio weights $\mathbf{w}$ can now be solved as:

$$\mathbf{w} = \frac{1}{2}(\lambda_1 \mathbb{C}^{-1}\mathbb{1} + \lambda_2 \mathbb{C}^{-1}\bar{\mathbf{r}}) =$$

$$\frac{1}{2}\begin{bmatrix} \mathbb{C}^{-1}\mathbb{1} \\ \mathbb{C}^{-1}\bar{\mathbf{r}} \end{bmatrix}^T \lambda = \begin{bmatrix} \mathbb{C}^{-1}\mathbb{1} \\ \mathbb{C}^{-1}\bar{\mathbf{r}} \end{bmatrix}^T \mathbb{F}^{-1} u =$$

$$\frac{1}{\det \mathbb{F}}\begin{bmatrix} \mathbb{C}^{-1}\mathbb{1} \\ \mathbb{C}^{-1}\bar{\mathbf{r}} \end{bmatrix}^T \begin{bmatrix} c_{rr} & -c_{r1} \\ -c_{r1} & c_{11} \end{bmatrix} \begin{bmatrix} 1 \\ r_t \end{bmatrix} =$$

$$\frac{(c_{rr} - c_{r1} r_t)\, \mathbb{C}^{-1}\mathbb{1} + (c_{11} r_t - c_{r1})\, \mathbb{C}^{-1}\bar{\mathbf{r}}}{\det \mathbb{F}}$$

With $c_{11} = \mathbb{1}^T \mathbb{C}^{-1} \mathbb{1}$, $c_{r1} = \mathbb{1}^T \mathbb{C}^{-1} \bar{\mathbf{r}}$, $c_{rr} = \bar{\mathbf{r}}^T \mathbb{C}^{-1} \bar{\mathbf{r}}$. And $\det \mathbb{F} = c_{11} c_{rr} - c_{r1}^2$ is the determinant of the matrix $\mathbb{F}$.

The above formula shows that the efficient portfolio weights are a linear function of the target return.

Therefore a convex sum of two efficient portfolio weights: $w = \alpha w_1 + (1 - \alpha) w_2$, are also the weights of an *efficient portfolio*, with target return equal to: $r_t = \alpha r_1 + (1 - \alpha) r_2$

```
> # Calculate vector of mean returns
> retm <- colMeans(retp)
> # Specify the target return
> retarg <- 1.5*mean(retp)
> # Products of inverse with mean returns and unit vector
> c11 <- drop(t(unitv) %*% covinv %*% unitv)
> cr1 <- drop(t(unitv) %*% covinv %*% retm)
> crr <- drop(t(retm) %*% covinv %*% retm)
> fmat <- matrix(c(c11, cr1, cr1, crr), nc=2)
> # Solve for the Lagrange multipliers
> lagm <- solve(a=fmat, b=c(2, 2*retarg))
> # Calculate the efficient portfolio weights
> weightv <- 0.5*drop(covinv %*% cbind(unitv, retm) %*% lagm)
> # Calculate constraints
> all.equal(1, sum(weightv))
> all.equal(retarg, sum(retm*weightv))
```

Variance of the *Efficient Portfolios*

The *efficient portfolio* variance is equal to:

$$\sigma^2 = \mathbf{w}^T \mathbb{C} \, \mathbf{w} = \frac{1}{4} \lambda^T \mathbb{F} \lambda = u^T \mathbb{F}^{-1} u =$$
$$\frac{1}{\det \mathbb{F}} \begin{bmatrix} 1 \\ r_t \end{bmatrix}^T \begin{bmatrix} c_{rr} & -c_{r1} \\ -c_{r1} & c_{11} \end{bmatrix} \begin{bmatrix} 1 \\ r_t \end{bmatrix} =$$
$$\frac{c_{11} r_t^2 - 2 c_{r1} r_t + c_{rr}}{\det \mathbb{F}}$$

The above formula shows that the variance of the *efficient portfolios* is a *parabola* with respect to the target return r_t.

The vertex of the *parabola* is the minimum variance portfolio: $r_{mv} = c_{r1} / c_{11} = \mathbb{1}^T \mathbb{C}^{-1} \bar{\mathbf{r}} / \mathbb{1}^T \mathbb{C}^{-1} \mathbb{1}$ and $\sigma^2_{mv} = 1/c_{11} = 1/\mathbb{1}^T \mathbb{C}^{-1} \mathbb{1}$.

The *efficient portfolio* variance can be expressed in terms of the difference $\Delta_r = r_t - r_{mv}$ as:

$$\sigma^2 = \frac{\Delta_r^2 + \det \mathbb{F}}{c_{11} \det \mathbb{F}}$$

So that if $\Delta_r = 0$ then $\sigma^2 = 1/c_{11}$.

Where $\det \mathbb{F} = c_{11} c_{rr} - c_{r1}^2$ is the determinant of the matrix $\mathbb{F}$.

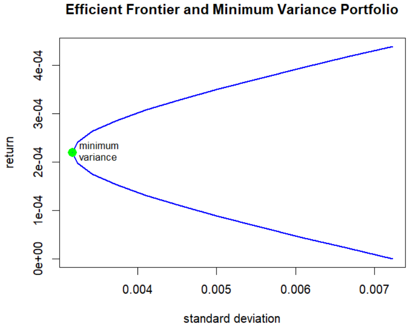
```
> # Calculate the efficient portfolio returns
> reteff <- drop(retp %*% weightv)
> reteffm <- mean(reteff)
> all.equal(reteffm, retarg)
> # Calculate the efficient portfolio variance in three ways
> uu <- c(1, retarg)
> finv <- solve(fmat)
> detf <- (c11*crr-cr1^2)  ##  det(fmat)
> all.equal(var(reteff),
+   drop(t(uu) %*% finv %*% uu),
+   (c11*reteffm^2-2*cr1*reteffm+crr)/detf)
```

The *Efficient Frontier*

The *efficient frontier* is the set of *efficient portfolios*, that have the lowest risk (standard deviation) for the given level of return.

The *efficient frontier* is the plot of the target returns r_t and the standard deviations of the *efficient portfolios*, which is a *hyperbola*.

```
> # Calculate the daily and mean minvar portfolio returns
> c11 <- drop(t(unitv) %*% covinv %*% unitv)
> weightv <- drop(covinv %*% unitv/c11)
> retmv <- (retp %*% weightv)
> retmvm <- sum(weightv*retm)
> # Calculate the minimum variance
> varmv <- 1/c11
> stdevmv <- sqrt(varmv)
> # Calculate efficient frontier from target returns
> retargv <- retmvm*(1+seq(from=(-1), to=1, by=0.1))
> stdevs <- sapply(retargv, function(rett) {
+   uu <- c(1, rett)
+   sqrt(drop(t(uu) %*% finv %*% uu))
+ })  ##  end sapply
```




```
> # Plot the efficient frontier
> plot(x=stdevs, y=retargv, t="l", col="blue", lwd=2,
+      main="Efficient Frontier and Minimum Variance Portfolio",
+      xlab="standard deviation", ylab="return")
> points(x=stdevmv, y=retmvm, col="green", lwd=6)
> text(x=stdevmv, y=retmvm, labels="minimum \nvariance",
+      pos=4, cex=0.8)
```

The *Tangent Line* and the *Risk-free* Rate

The *tangent* line connects the risk-free point $(\sigma = 0, r = r_f)$ with a single tangent point on the *efficient frontier*.

A *tangent* line can be drawn at every point on the *efficient frontier*.

The slope β of the *tangent* line can be calculated by differentiating the efficient portfolio variance σ^2 by the target return r_t:

$$\frac{d\sigma^2}{dr_t} = 2\sigma \frac{d\sigma}{dr_t} = \frac{2c_{11} r_t - 2c_{r1}}{\det \mathbb{F}}$$

$$\frac{d\sigma}{dr_t} = \frac{c_{11} r_t - c_{r1}}{\sigma \det \mathbb{F}}$$

$$\beta = \frac{\sigma \det \mathbb{F}}{c_{11} r_t - c_{r1}}$$

The *tangent* line connects the *tangent* point on the *efficient frontier* with a *risk-free* rate r_f.

The *risk-free* rate r_f can be calculated as the intercept of the tangent line:

$$r_f = r_t - \sigma \beta = r_t - \frac{\sigma^2 \det \mathbb{F}}{c_{11} r_t - c_{r1}} =$$

$$r_t - \frac{c_{11} r_t^2 - 2c_{r1} r_t + c_{rr}}{\det \mathbb{F}} \frac{\det \mathbb{F}}{c_{11} r_t - c_{r1}} =$$

$$r_t - \frac{c_{11} r_t^2 - 2c_{r1} r_t + c_{rr}}{c_{11} r_t - c_{r1}} = \frac{c_{r1} r_t - c_{rr}}{c_{11} r_t - c_{r1}}$$

```
> # Calculate standard deviation of efficient portfolio
> uu <- c(1, retarg)
> stdeveff <- sqrt(drop(t(uu) %*% finv %*% uu))
> # Calculate the slope of the tangent line
> detf <- (c11*crr-cr1^2)  ##  det(fmat)
> sharper <- (stdeveff*detf)/(c11*retarg-cr1)
> # Calculate the risk-free rate as intercept of the tangent line
> raterf <- retarg - sharper*stdeveff
> # Calculate the risk-free rate from target return
> all.equal(raterf,
+   (retarg*cr1-crr)/(retarg*c11-cr1))
```

The *Capital Market Line*

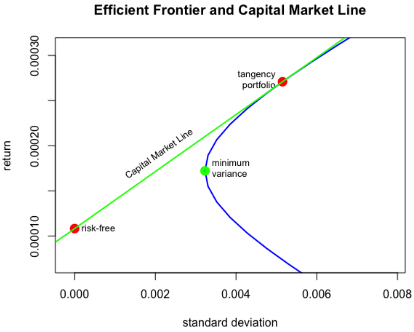
The *Capital Market Line* (CML) is the tangent line connecting the risk-free point ($\sigma = 0, r = r_f$) with a single tangent point on the *efficient frontier*.

The *tangency portfolio* is the *efficient portfolio* at the tangent point corresponding to the given *risk-free* rate.

Each value of the *risk-free* rate r_f corresponds to a unique *tangency portfolio*.

For a given *risk-free* rate r_f, the *tangency portfolio* has the highest *Sharpe ratio* among all the *efficient portfolios*.

```
> # Plot efficient frontier
> aspectr <- 1.0*max(stdevs)/diff(range(retargv)) ##  Aspect ratio
> plot(x=stdevs, y=retargv, t="l", col="blue", lwd=2, asp=aspectr,
+      xlim=c(0.4, 0.6)*max(stdevs), ylim=c(0.2, 0.9)*max(retargv),
+      main="Efficient Frontier and Capital Market Line",
+      xlab="standard deviation", ylab="return")
> # Plot the minimum variance portfolio
> points(x=stdevmv, y=retmvm, col="green", lwd=6)
> text(x=stdevmv, y=retmvm, labels="minimum \nvariance",
+      pos=4, cex=0.8)
```




```
> # Plot the tangent portfolio
> points(x=stdeveff, y=retarg, col="red", lwd=6)
> text(x=stdeveff, y=retarg, labels="tangency\nportfolio", pos=2, ce
> # Plot the risk-free point
> points(x=0, y=raterf, col="red", lwd=6)
> text(x=0, y=raterf, labels="risk-free", pos=4, cex=0.8)
> # Plot the tangent line
> abline(a=raterf, b=sharper, lwd=2, col="green")
> text(x=0.6*stdev, y=0.8*retarg,
+      labels="Capital Market Line", pos=2, cex=0.8,
+      srt=180/pi*atan(aspectr*sharper))
```

The *Capital Market Line* Portfolios

The points on the *Capital Market Line* represent portfolios consisting of the *tangency portfolio* and the *risk-free* asset (bond).

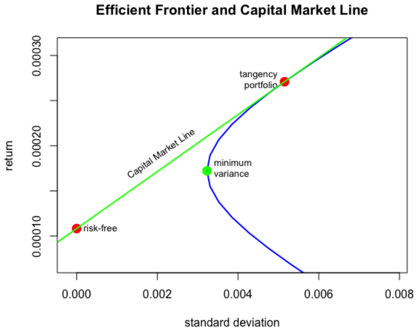
The *Capital Market Line* represents delevered and levered portfolios, consisting of the *tangency portfolio* combined with the *risk-free* asset (bond).

The *CML* portfolios have weights proportional to the tangency portfolio weights.

The *CML* portfolios above the tangent point are levered with respect to the *tangency portfolio* through borrowing at the *risk-free* rate r_f. Their weights are equal to the tangency portfolio weights multiplied by a factor greater than 1.

The *CML* portfolios below the tangent point are delevered with respect to the *tangency portfolio* through investing at the *risk-free* rate r_f. Their weights are equal to the tangency portfolio weights multiplied by a factor less than 1.

All the *CML* portfolios have the same *Sharpe ratio*.



Maximum Sharpe Portfolio Weights

The *Sharpe ratio* is equal to the ratio of excess returns divided by the portfolio standard deviation:

$$SR = \frac{\mathbf{w}^T \mu}{\sigma}$$

Where $\mu = \bar{\mathbf{r}} - r_f$ is the vector of mean excess returns (in excess of the risk-free rate r_f), $\mathbf{w}$ is the vector of portfolio weights, and $\sigma = \sqrt{\mathbf{w}^T \mathbb{C}\, \mathbf{w}}$, where $\mathbb{C}$ is the covariance matrix of returns.

We can calculate the *maximum Sharpe* portfolio weights by setting the derivative of the *Sharpe ratio* with respect to the weights $\mathbf{w}$ to zero:

$$d_w SR = \frac{1}{\sigma}(\mu^T - \frac{(\mathbf{w}^T \mu)(\mathbf{w}^T \mathbb{C})}{\sigma^2}) = 0$$

We then get:

$$(\mathbf{w}^T \mathbb{C}\, \mathbf{w})\, \mu = (\mathbf{w}^T \mu)\, \mathbb{C}\mathbf{w}$$

We can multiply the above equation by $\mathbb{C}^{-1}$ to get:

$$\mathbf{w} = \frac{\mathbf{w}^T \mathbb{C}\, \mathbf{w}}{\mathbf{w}^T \mu}\, \mathbb{C}^{-1} \mu$$

We can finally rescale the weights so that they satisfy the linear constraint $\mathbf{w}^T \mathbb{1} = 1$:

$$\mathbf{w} = \frac{\mathbb{C}^{-1} \mu}{\mathbb{1}^T \mathbb{C}^{-1} \mu}$$

These are the weights of the *maximum Sharpe* portfolio, with the vector of mean excess returns equal to μ, and the covariance matrix equal to $\mathbb{C}$.

The *maximum Sharpe* portfolio is an *efficient portfolio*, and so its mean return is equal to some target return r_t: $\bar{\mathbf{r}}^T \mathbf{w} = \sum_{i=1}^{n} w_i r_i = r_t$.

The mean return of the *maximum Sharpe* portfolio is equal to:

$$r_t = \bar{\mathbf{r}}^T \mathbf{w} = \frac{\bar{\mathbf{r}}^T \mathbb{C}^{-1} \mu}{\mathbb{1}^T \mathbb{C}^{-1} \mu} = \frac{\bar{\mathbf{r}}^T \mathbb{C}^{-1} (\bar{\mathbf{r}} - r_f)}{\mathbb{1}^T \mathbb{C}^{-1} (\bar{\mathbf{r}} - r_f)} =$$

$$\frac{\bar{\mathbf{r}}^T \mathbb{C}^{-1} \mathbb{1}\, r_f - \bar{\mathbf{r}}^T \mathbb{C}^{-1} \mathbb{1}\, \bar{\mathbf{r}}}{\mathbb{1}^T \mathbb{C}^{-1} \mathbb{1}\, r_f - \bar{\mathbf{r}}^T \mathbb{C}^{-1} \mathbb{1}} = \frac{c_{r1}\, r_f - c_{rr}}{c_{11}\, r_f - c_{r1}}$$

The above formula calculates the target return r_t from the risk-free rate r_f.

Returns and Variance of the *Maximum Sharpe* Portfolio

The *maximum Sharpe* portfolio weights depend on the value of the risk-free rate r_f:

$$\mathbf{w} = \frac{\mathbb{C}^{-1}\mu}{\mathbb{1}^T\mathbb{C}^{-1}\mu} = \frac{\mathbb{C}^{-1}(\bar{\mathbf{r}} - r_f)}{\mathbb{1}^T\mathbb{C}^{-1}(\bar{\mathbf{r}} - r_f)}$$

The mean return of the *maximum Sharpe* portfolio is equal to:

$$r_t = \bar{\mathbf{r}}^T\mathbf{w} = \frac{\bar{\mathbf{r}}^T\mathbb{C}^{-1}\mu}{\mathbb{1}^T\mathbb{C}^{-1}\mu} = \frac{c_{r1}\, r_f - c_{rr}}{c_{11}\, r_f - c_{r1}}$$

The variance of the *maximum Sharpe* portfolio is equal to:

$$\sigma^2 = \mathbf{w}^T\mathbb{C}\,\mathbf{w} = \frac{\mu^T\mathbb{C}^{-1}\mathbb{C}\,\mathbb{C}^{-1}\mu}{(\mathbb{1}^T\mathbb{C}^{-1}\mu)^2} = \frac{\mu^T\mathbb{C}^{-1}\mu}{(\mathbb{1}^T\mathbb{C}^{-1}\mu)^2} =$$

$$\frac{(\bar{\mathbf{r}} - r_f)^T\mathbb{C}^{-1}(\bar{\mathbf{r}} - r_f)}{(\mathbb{1}^T\mathbb{C}^{-1}(\bar{\mathbf{r}} - r_f))^2} = \frac{c_{11}r_t^2 - 2c_{r1}r_t + c_{rr}}{\det\mathbb{F}}$$

The above formula expresses the *maximum Sharpe* portfolio variance as a function of its mean return r_t.

The *maximum Sharpe* ratio is equal to:

$$SR = \frac{\mathbf{w}^T\mu}{\sigma} = \frac{\mu^T\mathbb{C}^{-1}\mu}{\mathbb{1}^T\mathbb{C}^{-1}\mu} / \frac{\sqrt{\mu^T\mathbb{C}^{-1}\mu}}{\mathbb{1}^T\mathbb{C}^{-1}\mu} =$$

$$\sqrt{\mu^T\mathbb{C}^{-1}\mu} = \sqrt{(\bar{\mathbf{r}} - r_f)^T\mathbb{C}^{-1}(\bar{\mathbf{r}} - r_f)}$$

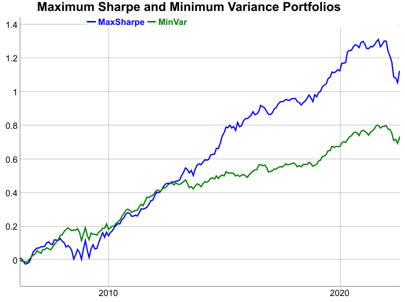
```
> # Calculate the mean excess returns
> raterf <- retarg - sharper*stdeveff
> retx <- (retm - raterf)
> # Calculate the efficient portfolio weights
> weightv <- 0.5*drop(covinv %*% cbind(unitv, retm) %*% lagm)
> # Calculate the maximum Sharpe weights
> weightms <- drop(covinv %*% retx)/sum(covinv %*% retx)
> all.equal(weightv, weightms)
> # Calculate the maximum Sharpe mean return in two ways
> all.equal(sum(retm*weightv), (cr1*raterf-crr)/(c11*raterf-cr1))
> # Calculate the maximum Sharpe daily returns
> retd <- (retp %*% weightms)
> # Calculate the maximum Sharpe variance in four ways
> detf <- (c11*crr-cr1^2)  ##  det(fmat)
> all.equal(var(retd),
+   t(weightv) %*% covmat %*% weightv,
+   (t(retx) %*% covinv %*% retx)/sum(covinv %*% retx)^2,
+   (c11*retarg^2-2*cr1*retarg+crr)/detf)
> # Calculate the maximum Sharpe ratio
> sqrt(252)*sum(weightv*retx)/
+   sqrt(drop(t(weightv) %*% covmat %*% weightv))
> # Calculate the stock Sharpe ratios
> sqrt(252)*sapply((retp - raterf), function(x) mean(x)/sd(x))
```

Maximum Sharpe and *Minimum Variance* Performance

The *maximum Sharpe* and *Minimum Variance* portfolios are both *efficient portfolios*, with the lowest risk (standard deviation) for the given level of return.

The *maximum Sharpe* portfolio has both a higher Sharpe ratio and higher absolute returns.

```
> # Calculate optimal portfolio returns
> wealthv <- cbind(retp %*% weightms, retp %*% weightmv)
> wealthv <- xts::xts(wealthv, zoo::index(retp))
> colnames(wealthv) <- c("MaxSharpe", "MinVar")
> # Calculate the Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv, function(x)
+   (mean(x)-raterf)/c(Sharpe=sd(x), Sortino=sd(x[x<0])))
> # Plot the log wealth
> endw <- rutils::calc_endpoints(wealthv, interval="weeks")
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Maximum Sharpe and Minimum Variance Portfolios") %>%
+   dyOptions(colors=c("blue", "green"), strokeWidth=2) %>%
+   dyLegend(show="always", width=500)
```



Maximum Sharpe Portfolio of Two Stocks

Given the covariance matrix of two stocks:

$$\mathbb{C} = \begin{bmatrix} \sigma^2_1 & \sigma_{12} \\ \sigma_{12} & \sigma^2_2 \end{bmatrix}$$

The inverse of the covariance matrix is equal to:

$$\mathbb{C}^{-1} = \frac{1}{(\sigma^2_1 \sigma^2_2 - \sigma^2_{12})} \begin{bmatrix} \sigma^2_2 & -\sigma_{12} \\ -\sigma_{12} & \sigma^2_1 \end{bmatrix}$$

Where σ^2_1 and σ^2_2 are the stock variances and σ_{12} is their covariance.

The *maximum Sharpe* portfolio weights are equal to:

$$\mathbf{w} = \frac{\mathbb{C}^{-1}\mu}{\mathbb{1}^T \mathbb{C}^{-1}\mu} = \frac{1}{D} \begin{bmatrix} \sigma^2_2 \mu_1 - \sigma_{12}\mu_2 \\ -\sigma_{12}\mu_1 + \sigma^2_1 \mu_2 \end{bmatrix}$$

Where μ_1 and μ_2 are the mean excess returns of the two stocks, and $D = \sigma^2_2 \mu_1 + \sigma^2_1 \mu_2 - \sigma_{12}(\mu_1 + \mu_2)$.

```
> # Calculate the covariance, variances and the mean returns
> retp <- na.omit(rutils::etfenv$returns[, c("VTI", "TLT")])
> retew <- 0.4*retp[, 1] + 0.6*retp[, 2]
> covm <- cov(retp)
> covv <- covm[1, 2]
> var1 <- covm[1, 1]
> var2 <- covm[2, 2]
> ret1 <- mean(retp[, 1])
> ret2 <- mean(retp[, 2])
> # Calculate the inverse covariance matrix
> scalev <- (var1*var2 - covv^2)
> covinv <- matrix(c(var2, -covv, -covv, var1), nrow=2)/scalev
> round(covinv %*% covm, 4)
> # Calculate the maximum Sharpe portfolio weights
> wv <- (covinv %*% c(ret1, ret2))
> # Or
> (var2*ret1 - covv*ret2)/scalev
> (-covv*ret1 + var1*ret2)/scalev
> # Calculate the maximum Sharpe portfolio pnls
> pnls <- retp %*% wv
> pnls <- sd(retew)/sd(pnls)*pnls
> # Calculate the Sharpe ratios
> wealthv <- cbind(retew, pnls)
> colnames(wealthv) <- c("Balanced", "MaxSharpe")
> # Calculate the Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Plot of combined log wealth
> endw <- rutils::calc_endpoints(wealthv, interval="weeks")
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Wealth of Fixed Share and Equal Wealth Portfolios") %>%
+   dyOptions(colors=c("blue", "red"), strokeWidth=2) %>%
+   dyLegend(show="always", width=300)
```

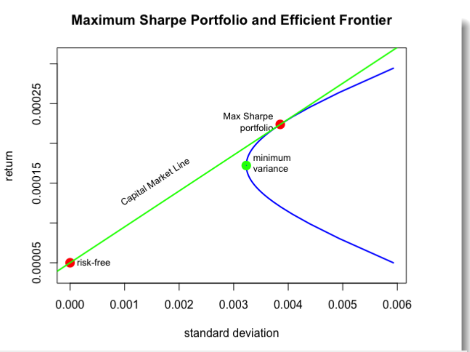
The *Maximum Sharpe* Portfolios and the *Efficient Frontier*

The *maximum Sharpe* portfolios are *efficient portfolios*, so they form the *efficient frontier*.

A *market portfolio* is the portfolio of all the available assets, with weights proportional to their market capitalizations.

The *maximum Sharpe* portfolio is sometimes considered to be the *market portfolio*, because it's the optimal portfolio for the given value of the risk-free rate r_f.

```
> # Calculate the maximum Sharpe portfolios for different risk-free
> detf <- (c11*crr-cr1^2)  ##  det(fmat)
> raterfv <- retmvm*seq(from=1.3, to=20, by=0.1)
> raterfv <- c(raterfv, retmvm*seq(from=(-20), to=0.7, by=0.1))
> effront <- sapply(raterfv, function(raterf) {
+   ##  Calculate the maximum Sharpe mean return
+   reteffm <- (cr1*raterf-crr)/(c11*raterf-cr1)
+   ##  Calculate the maximum Sharpe standard deviation
+   stdev <- sqrt((c11*reteffm^2-2*cr1*reteffm+crr)/detf)
+   c(return=reteffm, stdev=stdev)
+ })  ##  end sapply
> effront <- effront[, order(effront["return", ])]
> # Plot the efficient frontier
> reteffv <- effront["return", ]
> stdevs <- effront["stdev", ]
> aspectr <- 0.6*max(stdevs)/diff(range(reteffv)) ##  Aspect ratio
> plot(x=stdevs, y=reteffv, t="l", col="blue", lwd=2, asp=aspectr,
+   main="Maximum Sharpe Portfolio and Efficient Frontier",
+   xlim=c(0.001, max(stdevs)), xlab="standard deviation", ylab="r
> # Plot the minimum variance portfolio
> points(x=stdevmv, y=retmvm, col="green", lwd=6)
> text(x=stdevmv, y=retmvm, labels="minimum \nvariance", pos=4, ce
```




```
> # Calculate the maximum Sharpe return and standard deviation
> raterf <- min(reteffv)
> retmax <- (cr1*raterf-crr)/(c11*raterf-cr1)
> stdevmax <- sqrt((c11*retmax^2-2*cr1*retmax+crr)/detf)
> # Plot the maximum Sharpe portfolio
> points(x=stdevmax, y=retmax, col="red", lwd=6)
> text(x=stdevmax, y=retmax, labels="Max Sharpe\nportfolio", pos=2,
> # Plot the risk-free point
> points(x=0, y=raterf, col="red", lwd=6)
> text(x=0, y=raterf, labels="risk-free", pos=4, cex=0.8)
> # Plot the tangent line
> sharper <- (stdevmax*detf)/(c11*retmax-cr1)
> abline(a=raterf, b=sharper, lwd=2, col="green")
> text(x=0.6*stdevmax, y=0.8*retmax, labels="Capital Market Line",
+      pos=2, cex=0.8, srt=180/pi*atan(aspectr*sharper))
```

Efficient Portfolios and Their Tangent Lines

The *efficient frontier* consists of all the *maximum Sharpe* portfolios corresponding to different values of the risk-free rate.

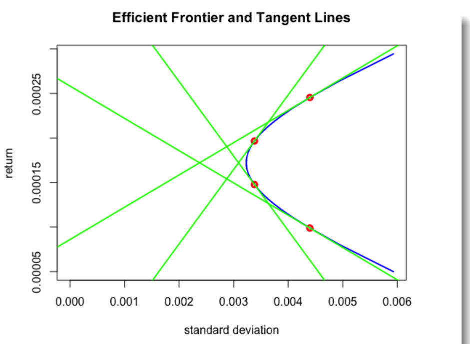
The target return can be expressed as a function of the risk-free rate as:

$$r_t = \frac{c_{r1}\, r_f - c_{rr}}{c_{11}\, r_f - c_{r1}}$$

If $r_f \to \pm\infty$ then $r_t \to r_{mv} = c_{r1}/c_{11}$.

But if the risk-free rate tends to the mean returns of the minimum variance portfolio: $r_f \to r_{mv} = c_{r1}/c_{11}$, then $r_t \to \pm\infty$, which means that there is no efficient portfolio corresponding to the risk-free rate equal to the mean returns of the minimum variance portfolio: $r_f = r_{mv} = c_{r1}/c_{11}$.

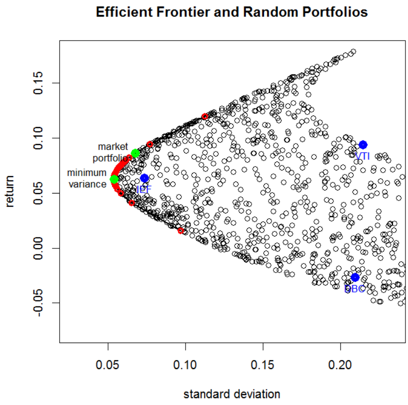
```
> # Plot the efficient frontier
> reteffv <- effront["return", ]
> stdevs <- effront["stdev", ]
> plot(x=stdevs, y=reteffv, t="l", col="blue", lwd=2,
+   xlim=c(0.0, max(stdevs)),
+   main="Efficient Frontier and Tangent Lines",
+   xlab="standard deviation", ylab="return")
```




```
> # Calculate vector of mean returns
> reteffv <- min(reteffv) + diff(range(reteffv))*c(0.2, 0.4, 0.6, 0.
> # Plot the tangent lines
> for (reteffm in reteffv) {
+   ##  Calculate the maximum Sharpe standard deviation
+   stdev <- sqrt((c11*reteffm^2-2*cr1*reteffm+crr)/detf)
+   ##  Calculate the slope of the tangent line
+   sharper <- (stdev*detf)/(c11*reteffm-cr1)
+   ##  Calculate the risk-free rate as intercept of the tangent li
+   raterf <- reteffm - sharper*stdev
+   ##  Plot the tangent portfolio
+   points(x=stdev, y=reteffm, col="red", lwd=3)
+   ##  Plot the tangent line
+   abline(a=raterf, b=sharper, lwd=2, col="green")
+ } ##  end for
```

Random Portfolios

```
> # Calculate random portfolios
> nportf <- 1000
> randportf <- sapply(1:nportf, function(it) {
+   weightv <- runif(nstocks-1, min=-0.25, max=1.0)
+   weightv <- c(weightv, 1-sum(weightv))
+   ##  Portfolio returns and standard deviation
+   c(return=sum(weightv*retm),
+     stdev=sqrt(drop(weightv %*% covmat %*% weightv)))
+ })  ##  end sapply
> # Plot scatterplot of random portfolios
> stdev <- min(effront["stdev", ])
> plot(x=randportf["stdev", ], y=randportf["return", ],
+      main="Efficient Frontier and Random Portfolios",
+      xlim=c(0.5*stdev, 0.8*max(randportf["stdev", ])),
+      xlab="standard deviation", ylab="return")
> # Plot maximum Sharpe portfolios
> lines(x=effront["stdev", ], y=effront["return", ], lwd=2)
> points(x=effront["stdev", ], y=effront["return", ],
+  col="red", lwd=3)
> # Plot the minimum variance portfolio
> points(x=stdevmv, y=retmvm, col="green", lwd=6)
> text(stdevmv, retmvm, labels="minimum\nvariance", pos=2, cex=0.8)
> # Plot efficient portfolio
> points(x=stdevmax, y=retmax, col="green", lwd=6)
> text(x=stdevmax, y=retmax, labels="market\nportfolio", pos=2, cex=
```

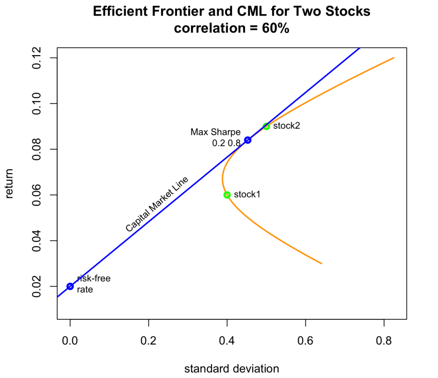



```
> # Plot individual assets
> points(x=sqrt(diag(covmat)), y=retm, col="blue", lwd=4)
> text(x=sqrt(diag(covmat)), y=retm, labels=names(retm),
+      lwd=4, col="blue", pos=1, cex=0.8)
```

Efficient Frontier for Two-stock Portfolio

The *efficient frontier* for a two-stock portfolio illustrates its dependence on the correlation and the risk-free rate.

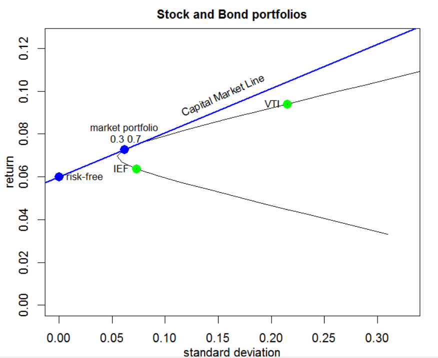
```
> # Define the parameters
> raterf <- 0.02 ##  Risk-free rate
> retp <- c(stock1=0.06, stock2=0.09) ##  Returns
> stdevs <- c(stock1=0.4, stock2=0.5) ##  Standard deviations
> corrp <- 0.6 ##  Correlation
> covmat <- matrix(c(1, corrp, corrp, 1), nc=2) ##  Covariance matri
> covmat <- t(t(stdevs*covmat)*stdevs)
> weightv <- seq(from=(-1), to=2, length.out=71) ##  Weights
> weightv <- cbind(weightv, 1-weightv)
> retport <- weightv %*% retp ##  Portfolio returns
> portfsd <- sqrt(rowSums(weightv*(weightv %*% covmat))) ##  Portfol
> sharper <- (retport-raterf)/portfsd ##  Portfolio Sharpe ratios
> # Plot the efficient frontier
> widthp <- 6; heightp <- 5
> plot(portfsd, retport, t="l",
+  main=paste0("Efficient Frontier and CML for Two Stocks\ncorrelat
+  xlab="standard deviation", ylab="return",
+  lwd=2, col="orange", xlim=c(0, max(portfsd)), ylim=c(0.01, max(re
> # Add the maximum Sharpe portfolio
> whichmax <- which.max(sharper)
> sharpem <- max(sharper) ##  Maximum Sharpe ratio
> retmax <- retport[whichmax]
> sdeff <- portfsd[whichmax]
> weightm <- round(weightv[whichmax], 2)
> points(sdeff, retmax, col="blue", lwd=3)
> text(x=sdeff, y=retmax, labels=paste(c("Max Sharpe\n",
+   structure(c(weightm, (1-weightm)), names=c("stock1", "stock2")
+   pos=2, cex=0.8)
```




```
> # Plot individual stocks
> points(stdevs, retp, col="green", lwd=3)
> text(stdevs, retp, labels=names(retp), pos=4, cex=0.8)
> # Add point at risk-free rate and draw Capital Market Line
> points(x=0, y=raterf, col="blue", lwd=3)
> text(0, raterf, labels="risk-free\nrate", pos=4, cex=0.8)
> abline(a=raterf, b=sharpem, lwd=2, col="blue")
> rangev <- par("usr")
> text(sdeff/2, (retmax+raterf)/2,
+      labels="Capital Market Line", cex=0.8, , pos=3,
+      srt=45*atan(sharpem*(rangev[2]-rangev[1])/
+              (rangev[4]-rangev[3])*heightp/widthp)/(0.25*pi))
```

Efficient Frontier of Stock and Bond Portfolios

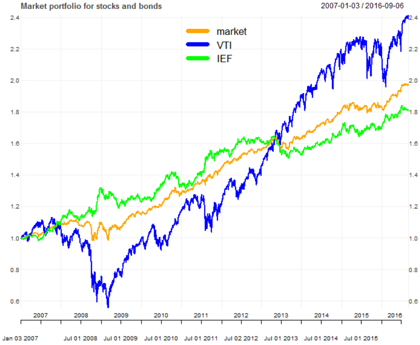
```
> # Vector of symbol names
> symbolv <- c("VTI", "IEF")
> # Matrix of portfolio weights
> weightv <- seq(from=(-1), to=2, length.out=31)
> weightv <- cbind(weightv, 1-weightv)
> # Calculate portfolio returns and volatilities
> retp <- na.omit(rutils::etfenv$returns[, symbolv])
> retport <- retp %*% t(weightv)
> portfv <- cbind(252*colMeans(retport),
+   sqrt(252)*matrixStats::colSds(retport))
> colnames(portfv) <- c("returns", "stdev")
> raterf <- 0.06
> portfv <- cbind(portfv,
+   (portfv[, "returns"]-raterf)/portfv[, "stdev"])
> colnames(portfv)[3] <- "Sharpe"
> whichmax <- which.max(portfv[, "Sharpe"])
> sharpem <- portfv[whichmax, "Sharpe"]
> plot(x=portfv[, "stdev"], y=portfv[, "returns"],
+      main="Stock and Bond portfolios", t="l",
+      xlim=c(0, 0.7*max(portfv[, "stdev"])), ylim=c(0, max(portfv[
+      xlab="standard deviation", ylab="return")
> # Add blue point for efficient portfolio
> points(x=portfv[whichmax, "stdev"], y=portfv[whichmax, "returns"]
> text(x=portfv[whichmax, "stdev"], y=portfv[whichmax, "returns"],
+      labels=paste(c("efficient portfolio\n",
+   structure(c(weightv[whichmax, 1], weightv[whichmax, 2]), names=
+      pos=3, cex=0.8)
```




```
> # Plot individual stocks
> retm <- 252*sapply(retport, mean)
> stdevs <- sqrt(252)*sapply(retport, sd)
> points(stdevs, retm, col="green", lwd=6)
> text(stdevs, retm, labels=names(retport), pos=2, cex=0.8)
> # Add point at risk-free rate and draw Capital Market Line
> points(x=0, y=raterf, col="blue", lwd=6)
> text(0, raterf, labels="risk-free", pos=4, cex=0.8)
> abline(a=raterf, b=sharpem, col="blue", lwd=2)
> rangev <- par("usr")
> text(max(portfv[, "stdev"])/3, 0.75*max(portfv[, "returns"]),
+      labels="Capital Market Line", cex=0.8, , pos=3,
+      srt=45*atan(sharpem*(rangev[2]-rangev[1])/
+              (rangev[4]-rangev[3])*
+              heightp/widthp)/(0.25*pi))
```

Performance of Efficient Portfolio for Stocks and Bonds

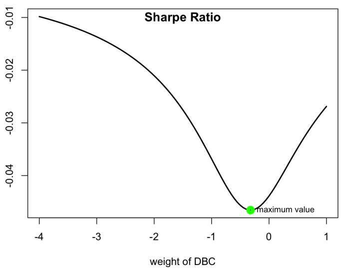
```
> # Calculate cumulative returns of VTI and IEF
> retsoptim <- lapply(retp, function(retp) exp(cumsum(retp)))
> retsoptim <- rutils::do_call(cbind, retsoptim)
> # Calculate the efficient portfolio returns
> retsoptim <- cbind(exp(cumsum(retp %*%
+     c(weightv[whichmax], 1-weightv[whichmax]))),
+   retsoptim)
> colnames(retsoptim)[1] <- "efficient"
> # Plot efficient portfolio with custom line colors
> themev <- chart_theme()
> themev$col$line.col <- c("orange", "blue", "green")
> chart_Series(retsoptim, theme=themev,
+    name="Efficient Portfolio for Stocks and Bonds")
> legend("top", legend=colnames(retsoptim),
+    cex=0.8, inset=0.1, bg="white", lty=1,
+    lwd=6, col=themev$col$line.col, bty="n")
```



Sharpe Ratio Objective Function

The function optimize() performs *one-dimensional* optimization over a single independent variable, in the specified interval.

```
> # Calculate daily ETF percentage returns
> symbolv <- c("VTI", "IEF", "DBC")
> nstocks <- NROW(symbolv)
> retp <- na.omit(rutils::etfenv$returns[, symbolv])
> nrows <- NROW(retp)
> # Create initial vector of portfolio weights
> weightv <- rep(1, NROW(symbolv))
> names(weightv) <- symbolv
> # Objective equal to minus Sharpe ratio
> objfun <- function(weightv, retp) {
+   retportf <- retp %*% weightv
+   stdev <- sd(retportf)
+   if (stdev == 0)
+     return(0)
+   else
+     return(-mean(retportf)/stdev)
+ }  ##  end objfun
> # Objective for equal weight portfolio
> objfun(weightv, retp=retp)
> optiml <- unlist(optimize(f=function(weightv)
+     objfun(c(1, 1, weightv), retp=retp),
+   interval=c(-10, 10)))
> # Vectorize objective function with respect to third weight
> objvec <- function(weightv) sapply(weightv,
+   function(weightv) objfun(c(1, 1, weightv), retp=retp))
> # Or
> objvec <- Vectorize(FUN=function(weightv)
+     objfun(c(1, 1, weightv), retp=retp),
+   vectorize.args="weightv")  ##  end Vectorize
> objvec(1)
> objvec(1:3)
```




```
> # Plot objective function with respect to third weight
> curve(expr=objvec, type="l", xlim=c(-4.0, 1.0),
+ xlab=paste("weight of", names(weightv[3])),
+ ylab="", lwd=2)
> title(main="Sharpe Ratio", line=(-1))  ##  Add title
> points(x=optiml[1], y=optiml[2], col="green", lwd=6)
> text(x=optiml[1], y=optiml[2],
+      labels="maximum value", pos=4, cex=0.8)
>
> ### Below is simplified plotting of objective function
> # Create vector of DBC weights
> weightv <- seq(from=-4, to=1, by=0.1)
> objv <- sapply(weightv, function(weightv)
+   objfun(c(1, 1, weightv), retp))
> plot(x=weightv, y=objv, t="l",
+ xlab="weight of DBC", ylab="", lwd=2)
> title(main="Sharpe Ratio", line=(-1))  ##  Add title
> points(x=optiml[1], y=optiml[2], col="green", lwd=6)
> text(x=optiml[1], y=optiml[2],
```

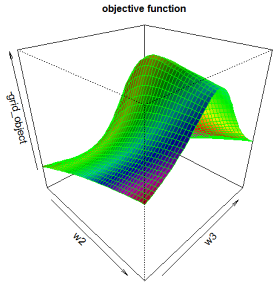
Perspective Plot of Portfolio Objective Function

The function persp() plots a 3d perspective surface plot of a function specified over a grid of argument values.

The function outer() calculates the values of a function over a grid spanned by two variables, and returns a matrix of function values.

The package *rgl* allows creating *interactive* 3d scatterplots and surface plots including perspective plots, based on the *OpenGL* framework.

```
> # Vectorize function with respect to two weights
> objvec <- Vectorize(
+   FUN=function(w1, w2, w3) objfun(c(w1, w2, w3), retp),
+   vectorize.args=c("w2", "w3"))  ##  end Vectorize
> # Calculate objective on 2-d (w2 x w3) parameter grid
> w2 <- seq(-3, 7, length=50)
> w3 <- seq(-5, 5, length=50)
> gridm <- outer(w2, w3, FUN=objvec, w1=1)
> rownames(gridm) <- round(w2, 2)
> colnames(gridm) <- round(w3, 2)
> # Perspective plot of objective function
> persp(w2, w3, -gridm,
+ theta=45, phi=30, shade=0.5,
+ col=rainbow(50), border="green",
+ main="objective function")
```




```
> # Interactive perspective plot of objective function
> library(rgl)
> rgl::persp3d(z=-gridm, zlab="objective",
+   col="green", main="objective function")
> rgl::persp3d(
+   x=function(w2, w3) {objvec(w1=1, w2, w3)},
+   xlim=c(-3, 7), ylim=c(-5, 5),
+   col="green", axes=FALSE)
> # Render the 3d surface plot of function
> rgl::rglwidget(elementId="plot3drgl", width=1000, height=1000)
```

Portfolio Optimization Using optim()

The functional optim() performs *multi-dimensional* optimization.

The argument par are the initial parameter values.

The argument fn is the objective function to be minimized.

The argument of the objective function which is to be optimized, must be a vector argument.

optim() accepts additional parameters bound to the dots "..." argument, and passes them to the fn objective function.

The arguments lower and upper specify the search range for the variables of the objective function fn.

method="L-BFGS-B" specifies the quasi-Newton optimization method.

The parameter control=list(factr=1e5) determines the precision of the L-BFGS-B method. A smaller factr value produces results with a higher precision, but it requires more iterations and a longer optimization. A larger factr value achieves a faster solution but less precise results.

optim() returns a list containing the location of the minimum and the objective function value.

```
> # Create initial vector of portfolio weights
> weightv <- rep(1, NROW(symbolv))
> names(weightv) <- symbolv
> # Optimization to find weights with maximum Sharpe ratio
> optiml <- optim(par=weightv,
+            fn=objfun,
+            retp=retp,
+            method="L-BFGS-B",
+            control=list(factr=1e5),
+            upper=c(10, 10, 10),
+            lower=c(-10, -10, -10))
> # Optimal parameters
> weightv <- optiml$par
> weightv <- weightv/sqrt(sum(weightv^2))
> weightv
> # Optimal Sharpe ratio
> -objfun(weightv, retp)
> # Calculate the weights from the inverse covariance matrix
> weightv <- drop(solve(cov(retp)) %*% sapply(retp, mean))
> weightv <- weightv/sqrt(sum(weightv^2))
> weightv
```

Optimization Algorithms Using `optim()`

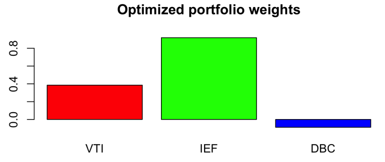
Table: Comparison of optimization methods for portfolio optimization

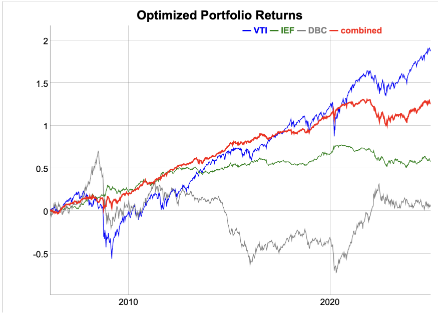
Method	**Type**	**Strengths**	**Weaknesses**	**Suitability for Portfolio Optimization**
BFGS	Quasi-Newton	Fast convergence, uses gradient + Hessian approximation	Memory-heavy for large problems	Strong choice for medium-sized portfolios
L-BFGS-B	Limited-memory BFGS with box constraints	Handles bounds (e.g., weights ≥ 0), memory-efficient	Requires smooth objective	Best overall for portfolio optimization with constraints
CG (Conjugate Gradient)	Gradient-based	Efficient for large-scale problems	Slower convergence than BFGS	Good for very large portfolios
Nelder-Mead	Derivative-free simplex	Works without gradients, robust for small problems	Slow, scales poorly in high dimensions	Limited use; not ideal for large portfolios
Brent	1D derivative-free	Very efficient in one dimension	Only works for single-variable problems	Not applicable to portfolios
SANN (Simulated Annealing)	Stochastic global search	Escapes local minima, works on non-smooth functions	Very slow, no guarantee of optimality	Useful for highly non-convex problems

Optimized Portfolio Returns

The optimized portfolio has both long and short positions, and has a larger *Sharpe ratio* than the individual assets.

```
> # barplot of optimal portfolio weights
> barplot(weightv, col=c("red", "green", "blue"),
+   main="Optimized portfolio weights")
> # Calculate the cumulative wealth of the optimized portfolio
> wealthv <- cbind(retp %*% weightv, retp)
> colnames(wealthv)[1] <- "combined"
> # Calculate the Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv, function(x)
+   (mean(x)-raterf)/c(Sharpe=sd(x), Sortino=sd(x[x<0])))
> # Plot the log wealth
> endw <- rutils::calc_endpoints(wealthv, interval="weeks")
> colv <- colnames(wealthv)
> colr <- c("red", "blue", "green", "grey")
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Optimized Portfolio Returns") %>%
+   dyOptions(colors=colr, strokeWidth=1) %>%
+   dySeries(name=colv[1], col="red", strokeWidth=2) %>%
+   dyLegend(show="always", width=300)
```





Package *DEoptim* for Global Optimization

The function DEoptim() from package *DEoptim* performs *global* optimization using the *Differential Evolution* algorithm.

Differential Evolution is a genetic algorithm which evolves a population of solutions over several generations,
https://link.springer.com/content/pdf/10.1023/A:1008202821328.pdf

The first generation of solutions is selected randomly.

Each new generation is obtained by combining solutions from the previous generation.

The best solutions are selected for creating the next generation.

The *Differential Evolution* algorithm is well suited for very large multi-dimensional optimization problems, such as portfolio optimization.

Gradient optimization methods are more efficient than *Differential Evolution* for smooth objective functions with no local minima.

```
> # Rastrigin function with vector argument for optimization
> rastrigin <- function(vecv, param=25){
+   sum(vecv^2 - param*cos(vecv))
+ }  ##  end rastrigin
> vecv <- c(pi/6, pi/6)
> rastrigin(vecv=vecv)
> library(DEoptim)
> # Optimize rastrigin using DEoptim
> optiml <- DEoptim::DEoptim(rastrigin,
+   upper=c(6, 6), lower=c(-6, -6),
+   DEoptim.control(trace=FALSE, itermax=50))
> # Optimal parameters and value
> optiml$optim$bestmem
> rastrigin(optiml$optim$bestmem)
> summary(optiml)
> plot(optiml)
```

Portfolio Optimization Using Package *Deoptim*

The *Differential Evolution* algorithm is well suited for very large multi-dimensional optimization problems, such as portfolio optimization.

With the parameter `control=list(parallelType=1)`, `DEoptim::DEoptim()` performs the optimization using parallel computing.

With the parameter `control=list(itermax=100)`, `DEoptim::DEoptim()` performs at most 100 iterations.

To achieve a good precision, `itermax` should be set to a much larger value, such as 10000. 100 iterations.

```
> # Perform optimization using DEoptim
> optiml <- DEoptim::DEoptim(fn=objfun,
+   upper=rep(10, NCOL(retp)),
+   lower=rep(-10, NCOL(retp)),
+   retp=retp,
+   control=list(trace=FALSE, itermax=100, parallelType=1))
> weightv <- optiml$optim$bestmem
> names(weightv) <- colnames(retp)
> weightv <- weightv/sqrt(sum(weightv^2))
> weightv
```

Efficient Portfolios With *Ridge Regularization*

Portfolio optimization has a very large number of parameters, equal to the number of stocks in the portfolio. This causes the *overfitting* of the stock portfolio weights to the in-sample returns.

The technique of *regularization* is designed to reduce the overfitting of the parameters in a model.

Ridge regularization adds a penalty term to the objective function, proportional to the sum of squares of the weights: $\mathbf{w}^T \mathbf{w} = \sum_{i=1}^{n} w_i^2$.

The Lagrangian for the efficient portfolio weights with *ridge regularization* is:

$$\mathcal{L} = \mathbf{w}^T \mathbb{C} \, \mathbf{w} - \lambda_1 (\mathbf{w}^T \mathbb{1} - 1) - \lambda_2 (\mathbf{w}^T \mathbf{r} - r_t) + \lambda \mathbf{w}^T \mathbf{w}$$

Where the regularization intensity λ determines the amount of regularization.

The Lagrangian can be rewritten using the ridge covariance matrix $\mathbb{C}_r = \mathbb{C} + \lambda \mathbb{1}$:

$$\mathcal{L} = \mathbf{w}^T \mathbb{C}_r \, \mathbf{w} - \lambda_1 (\mathbf{w}^T \mathbb{1} - 1) - \lambda_2 (\mathbf{w}^T \mathbf{r} - r_t)$$

This is the efficient portfolio Lagrangian with the ridge covariance matrix $\mathbb{C}_r$.

Ridge regularization adds a constant to the variances of the returns and reduces their correlations.

> *Ridge regularization* shrinks the covariance matrix towards the unit matrix, and shrinks the portfolio weights towards equal weights.

Portfolio Optimization With *Ridge Regularization*

Maximizing the efficient portfolio Lagrangian is equivalent to maximizing the *Sharpe ratio*:

$$SR = \frac{\mathbf{w}^T \mu}{\sigma}$$

Where $\mu = \mathbf{r} - r_f$ is the vector of excess returns, and $\sigma = \sqrt{\mathbf{w}^T \mathbb{C}_r \mathbf{w}}$ is the standard deviation of portfolio with the ridge covariance matrix $\mathbb{C}_r = \mathbb{C} + \lambda \mathbb{1}$.

The portfolio weights satisfying the linear constraint $\mathbf{w}^T \mathbb{1} = 1$ are:

$$\mathbf{w} = \frac{\mathbb{C}_r^{-1} \mu}{\mathbb{1}^T \mathbb{C}_r^{-1} \mu}$$

These are the weights of the *maximum Sharpe* portfolio, with the vector of mean excess returns equal to μ, and the covariance matrix equal to $\mathbb{C}_r$.

```
> # Objective equal to minus ridge Sharpe ratio
> objfun <- function(weightv, retp, covret, lambdaf=0, lambdap=1e3)
+   retportf <- retp %*% weightv
+   covmat <- covret + lambdaf*diag(ncol(retx))
+   stdev <- sqrt(weightv %*% covmat %*% weightv)
+   if (stdev == 0)
+     return(0)
+   else
+     return(-mean(retportf)/stdev + lambdap*(sum(weightv)-1)^2)
+ }  ##  end objfun
> # Calculate ridge covariance matrix of returns and its inverse
> covret <- cov(retx)
> covmat <- covret + lambdaf*diag(ncol(retx))
> covinv <- solve(a=covmat)
> # Create initial vector of portfolio weights
> weightv <- rep(1, NROW(symbolv))
> names(weightv) <- symbolv
> # Calculate vector of excess returns
> raterf <- 0.03/252
> retx <- (retp - raterf)
> # Calculate the weights using optimization
> lambdaf <- 300.0
> objfun(weightv, retp=retx, lambdaf=lambdaf)
> optiml <- optim(par=weightv,
+           fn=objfun,
+           retp=retx,
+           covret=covret, # Covariance matrix of returns
+           lambdaf=lambdaf, # Ridge regularization intensity
+           lambdap=1e3, # Penalty for sum of weights not equal to
+           method="L-BFGS-B",
+           upper=c(10, 10, 10),
+           lower=c(-10, -10, -10))
> weightv <- optiml$par
> names(weightv) <- symbolv
> # Calculate the maximum Sharpe weights
> retm <- colMeans(retx)
> weightms <- drop(covinv %*% retm)/sum(covinv %*% retm)
> all.equal(weightv, weightms)
```

Portfolio Optimization With *Elastic Net Regularization*

The technique of *regularization* is designed to reduce the number of parameters in a model, for example in portfolio optimization.

The *regularization* techniques often add a penalty term to the objective function.

The penalty term in *ridge* regularization is the sum of squares of the weights: $\sum_{i=1}^{n} w_i^2$. The *ridge* penalty shrinks (reduces) the weights together, without shrinking some of them to zero.

The penalty term in *LASSO* regularization is the sum of the absolute values of the weights: $\sum_{i=1}^{n} |w_i|$. The *LASSO* penalty shrinks the less important weights to zero, similar to dimension reduction.

The *elastic net* regularization is a combination of *ridge* regularization and *LASSO* regularization:

$$w_{max} = \underset{w}{\arg\max}[\frac{\mathbf{w}^T \mu}{\sigma} - \lambda((1-\alpha)\sum_{i=1}^{n} w_i^2 + \alpha \sum_{i=1}^{n} |w_i|)]$$

The λ parameter determines the amount of regularization, and α controls the balance between the *ridge* and *LASSO* penalties.

```
> # Objective with regularization penalty
> objfun <- function(weightv, retp, lambdaf, alphaf) {
+   retportf <- retp %*% weightv
+   stdev <- sd(retportf)
+   if (stdev == 0)
+     return(0)
+   else {
+     penaltyv <- lambdaf*((1-alphaf)*sum(weightv^2) +
+ alphaf*sum(abs(weightv)))
+     return(-mean(retportf)/stdev + penaltyv)
+   }  ##  end if
+ }  ##  end objfun
> # Objective for equal weight portfolio
> weightv <- rep(1, NROW(symbolv))
> names(weightv) <- symbolv
> lambdaf <- 0.5 ; alphaf <- 0.5
> objfun(weightv, retp=retp, lambdaf=lambdaf, alphaf=alphaf)
> # Perform optimization using DEoptim
> optiml <- DEoptim::DEoptim(fn=objfun,
+   upper=rep(10, NCOL(retp)),
+   lower=rep(-10, NCOL(retp)),
+   retp=retp,
+   lambdaf=lambdaf,
+   alphaf=alphaf,
+   control=list(trace=FALSE, itermax=100, parallelType=1))
> weightv <- optiml$optim$bestmem
> names(weightv) <- colnames(retp)
> weightv <- weightv/sqrt(sum(weightv^2))
> weightv
```

Optimal Stock Portfolio Weights

The portfolio weights obtained from inverting the covariance matrix are the same as the weights obtained from portfolio optimization.

```
> # Load stock returns
> load("/Users/jerzy/Develop/lecture_slides/data/sp500_returns.RDat
> datev <- zoo::index(na.omit(retstock$GOOGL))
> retp <- retstock[datev] ##  Subset the returns to GOOGL
> # Remove the stocks with any NA values
> numna <- sapply(retp, function(x) sum(is.na(x)))
> retp <- retp[, numna==0]
> # Select 100 random stocks
> retp <- retp[, sample(NCOL(retp), 100)]
> symbolv <- colnames(retp)
> datev <- zoo::index(retp)
> retis <- retp["/2014"] ##  In-sample returns
> raterf <- 0.03/252
> retx <- (retis - raterf) ##  Excess returns
> # Calculate the maximum Sharpe weights in-sample interval
> colmeanv <- colMeans(retx, na.rm=TRUE)
> covmat <- cov(retx, use="pairwise.complete.obs")
> invreg <- MASS::ginv(covmat)
> weightv <- drop(invreg %*% colmeanv)
> names(weightv) <- symbolv
```

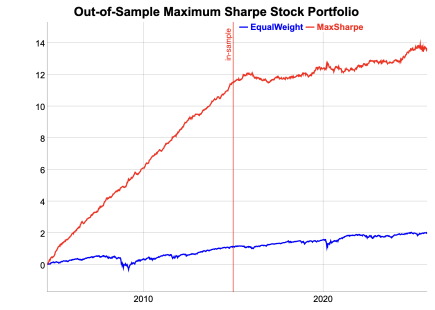
```
> # Calculate the weights using optimization without regularization
> lambdaf <- 0.0 ; alphaf <- 1.0
> optiml <- optim(par=weightv,
+           fn=objfun,
+           retp=retx,
+           covret=covret, # Covariance matrix of returns
+           lambdaf=lambdaf,
+           alphaf=alphaf,
+           method="L-BFGS-B",
+           control=list(factr=1e5),
+           upper=c(10, 10, 10),
+           lower=c(-10, -10, -10))
> weighto <- optiml$par
> names(weighto) <- symbolv
> all.equal(weighto, weightv)
```

Optimal Stock Portfolio Out-of-Sample

The *portfolio optimization* strategy for stocks is *overfit* in the *in-sample* interval.

Therefore the strategy performance is poor in the *out-of-sample* interval.

```
> # Load S&P 500 stock returns
> load("/Users/jerzy/Develop/lecture_slides/data/sp500_returns.RData
> datev <- zoo::index(na.omit(retstock$GOOGL))
> retp <- retstock[datev] ##  Subset the returns to GOOGL
> # Remove the stocks with any NA values
> numna <- sapply(retp, function(x) sum(is.na(x)))
> retp <- retp[, numna==0]
> # Select 100 random stocks
> retp <- retp[, sample(NCOL(retp), 100)]
> symbolv <- colnames(retp)
> datev <- zoo::index(retp)
> retis <- retp["/2014"] ##  In-sample returns
> raterf <- 0.03/252
> retx <- (retis - raterf) ##  Excess returns
> # Maximum Sharpe weights in-sample interval
> colmeanv <- colMeans(retx, na.rm=TRUE)
> covmat <- cov(retx, use="pairwise.complete.obs")
> invreg <- MASS::ginv(covmat)
> weightv <- drop(invreg %*% colmeanv)
> names(weightv) <- symbolv
> # Calculate the maximum Sharpe portfolio returns
> pnlmaxs <- HighFreq::mult_mat(weightv, retp)
> pnlmaxs <- rowMeans(pnlmaxs, na.rm=TRUE)
> pnlmaxs <- xts::xts(pnlmaxs, datev)
> retew <- xts::xts(rowMeans(retp, na.rm=TRUE), datev)
> pnlmaxs <- pnlmaxs*sd(retew["/2014"])/sd(pnlmaxs["/2014"])
> # Calculate the regularization portfolio returns
> lambdaf <- 1.0 ; alphaf <- 1.0
> optiml <- optim(par=weightv,
+           fn=objfun,
+           retp=retx,
```




```
> wealthv <- cbind(pnlmaxs, pnls)
> colnames(wealthv) <- c("MaxSharpe", "Shrink")
> # Calculate the in-sample Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv["/2014"], function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Calculate the out-of-sample Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv["2015/"], function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Plot of cumulative portfolio returns
> endw <- rutils::calc_endpoints(wealthv, interval="weeks")
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Out-of-Sample Maximum Sharpe Stock Portfolio") %>%
+   dyOptions(colors=c("blue", "red"), strokeWidth=2) %>%
+   dyEvent(end(retis[, 1]), label="in-sample", strokePattern="solid
+   dyLegend(width=300)
```

Optimal Portfolios Under Zero Correlation

If the correlations of returns are equal to zero, then the covariance matrix is diagonal:

$$\mathbb{C} = \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{pmatrix}$$

Where σ_i^2 is the variance of returns of asset `i`.

The inverse of $\mathbb{C}$ is then simply:

$$\mathbb{C}^{-1} = \begin{pmatrix} \sigma_1^{-2} & 0 & \cdots & 0 \\ 0 & \sigma_2^{-2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^{-2} \end{pmatrix}$$

The *minimum variance* portfolio weights are proportional to the inverse of the individual variances:

$$w_i = \frac{1}{\sigma_i^2 \sum_{i=1}^n \sigma_i^{-2}}$$

The *maximum Sharpe* portfolio weights are proportional to the ratio of the excess returns divided by the individual variances:

$$w_i = \frac{\mu_i}{\sigma_i^2 \sum_{i=1}^n \mu_i \sigma_i^{-2}}$$

The portfolio weights are proportional to the *Kelly ratios* - the excess returns divided by the variances:

$$w_i \propto \frac{\mu_i}{\sigma_i^2}$$

Maximum Return Portfolio Using Linear Programming

The stock weights of the maximum return portfolio are obtained by maximizing the portfolio returns:

$$w_{max} = \arg\max_{w} [\bar{\mathbf{r}}^T \mathbf{w}] = \arg\max_{w} [\sum_{i=1}^{n} w_i r_i]$$

Where $\mathbf{r}$ is the vector of returns, and $\mathbf{w}$ is the vector of portfolio weights, with a linear constraint:

$$\mathbf{w}^T \mathbb{1} = \sum_{i=1}^{n} w_i = 1$$

And a box constraint:

$$0 \leq w_i \leq 1$$

The weights of the maximum return portfolio can be calculated using linear programming (*LP*), which is the optimization of linear objective functions subject to linear constraints.

The function `Rglpk_solve_LP()` from package *Rglpk* solves linear programming problems by calling the *GNU Linear Programming Kit* library.

```
> library(rutils)
> library(Rglpk)
> # Vector of symbol names
> symbolv <- c("VTI", "IEF", "DBC")
> nstocks <- NROW(symbolv)
> # Calculate the objective vector - the mean returns
> retp <- na.omit(rutils::etfenv$returns[, symbolv])
> objvec <- colMeans(retp)
> # Specify matrix of linear constraint coefficients
> coeffm <- matrix(c(rep(1, nstocks), 1, 1, 0),
+            nc=nstocks, byrow=TRUE)
> # Specify the logical constraint operators
> logop <- c("==", "<=")
> # Specify the vector of constraints
> consv <- c(1, 0)
> # Specify box constraints (-1, 1) (default is c(0, Inf))
> boxc <- list(lower=list(ind=1:nstocks, val=rep(-1, nstocks)),
+        upper=list(ind=1:nstocks, val=rep(1, nstocks)))
> # Perform optimization
> optiml <- Rglpk::Rglpk_solve_LP(
+   obj=objvec,
+   mat=coeffm,
+   dir=logop,
+   rhs=consv,
+   bounds=boxc,
+   max=TRUE)
> all.equal(optiml$optimum, sum(objvec*optiml$solution))
> optiml$solution
> coeffm %*% optiml$solution
```

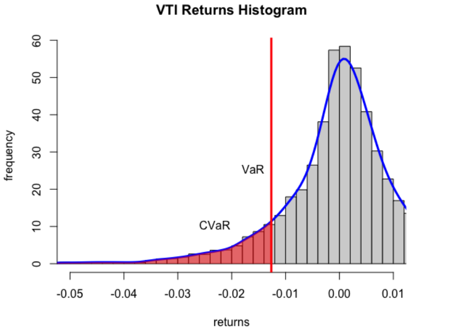
Conditional Value at Risk (*CVaR*)

The *Conditional Value at Risk* (*CVaR*) is equal to the average of the *VaR* for confidence levels less than a given confidence level α:

$$\mathrm{CVaR} = \frac{1}{\alpha} \int_0^{\alpha} \mathrm{VaR}(p)\, \mathrm{d}p$$

The *Conditional Value at Risk* is also called the Expected Shortfall (*ES*), or the Expected Tail Loss (*ETL*).

The function `density()` calculates a kernel estimate of the probability density for a sample of data, and returns a list with a vector of loss values and a vector of corresponding densities.




```
> # Calculate the VTI percentage returns
> retp <- na.omit(rutils::etfenv$returns$VTI)
> confl <- 0.1
> varisk <- quantile(retp, confl)
> cvar <- mean(retp[retp < varisk])
> # Or
> sortv <- sort(as.numeric(retp))
> varind <- round(confl*NROW(retp))
> varisk <- sortv[varind]
> cvar <- mean(sortv[1:varind])
> # Plot histogram of VTI returns
> varmin <- (-0.05)
> histp <- hist(retp, col="lightgrey",
+   xlab="returns", breaks=100, xlim=c(varmin, 0.01),
+   ylab="frequency", freq=FALSE, main="VTI Returns Histogram")
```

```
> # Plot density of losses
> densv <- density(retp, adjust=1.5)
> lines(densv, lwd=3, col="blue")
> # Add line for VaR
> abline(v=varisk, col="red", lwd=3)
> ymax <- max(densv$y)
> text(x=varisk, y=2*ymax/3, labels="VaR", lwd=2, pos=2)
> # Add shading for CVaR
> rangev <- (densv$x < varisk) & (densv$x > varmin)
> polygon(
+   c(varmin, densv$x[rangev], varisk),
+   c(0, densv$y[rangev], 0),
+   col=rgb(1, 0, 0,0.5), border=NA)
> text(x=1.5*varisk, y=ymax/7, labels="CVaR", lwd=2, pos=2)
```

CVaR Portfolio Weights Using Linear Programming

The stock weights of the minimum *CVaR* portfolio can be calculated using linear programming (*LP*), which is the optimization of linear objective functions subject to linear constraints,

$$w_{min} = \underset{w}{\arg\max} [\sum_{i=1}^{n} w_i b_i]$$

Where b_i is the negative objective vector, and $\mathbf{w}$ is the vector of portfolio weights, with a linear constraint:

$$\mathbf{w}^T \mathbb{1} = \sum_{i=1}^{n} w_i = 1$$

And a box constraint:

$$0 \leq w_i \leq 1$$

The function `Rglpk_solve_LP()` from package *Rglpk* solves linear programming problems by calling the *GNU Linear Programming Kit* library.

```
> library(rutils)  ##  Load rutils
> library(Rglpk)
> # Vector of symbol names and returns
> symbolv <- c("VTI", "IEF", "DBC")
> nstocks <- NROW(symbolv)
> retp <- na.omit(rutils::etfenv$returns[, symbolv])
> retm <- colMeans(retp)
> confl <- 0.05
> rmin <- 0 ; wmin <- 0 ; wmax <- 1
> weightsum <- 1
> ncols <- NCOL(retp) ##  number of assets
> nrows <- NROW(retp) ##  number of rows
> # Create objective vector
> objvec <- c(numeric(ncols), rep(-1/(confl/nrows), nrows), -1)
> # Specify matrix of linear constraint coefficients
> coeffm <- rbind(cbind(rbind(1, retm),
+                 matrix(data=0, nrow=2, ncol=(nrows+1))),
+           cbind(coredata(retp), diag(nrows), 1))
> # Specify the logical constraint operators
> logop <- c("==", ">=", rep(">=", nrows))
> # Specify the vector of constraints
> consv <- c(weightvum, rmin, rep(0, nrows))
> # Specify box constraints (wmin, wmax) (default is c(0, Inf))
> boxc <- list(lower=list(ind=1:ncols, val=rep(wmin, ncols)),
+       upper=list(ind=1:ncols, val=rep(wmax, ncols)))
> # Perform optimization
> optiml <- Rglpk_solve_LP(obj=objvec, mat=coeffm, dir=logop, rhs=co
> all.equal(optiml$optimum, sum(objvec*optiml$solution))
> coeffm %*% optiml$solution
> as.numeric(optiml$solution[1:ncols])
```

Package *quadprog* for Quadratic Programming

Quadratic programming (QP) is the optimization of quadratic objective functions subject to linear constraints.

Let $O(x)$ be an objective function that is quadratic with respect to a vector variable x:

$$O(x) = \frac{1}{2} x^T \mathbb{Q} x - d^T x$$

Where $\mathbb{Q}$ is a *positive definite* matrix ($x^T \mathbb{Q} x > 0$), and d is a vector.

An example of a *positive definite* matrix is the covariance matrix of linearly independent variables.

Let the linear constraints on the variable x be specified as:

$$\mathbb{A} x \geq b$$

Where $\mathbb{A}$ is a matrix, and b is a vector.

The function `solve.QP()` from package *quadprog* performs optimization of quadratic objective functions subject to linear constraints.

```
> library(quadprog)
> # Minimum variance weights without constraints
> optiml <- solve.QP(Dmat=2*covmat,
+             dvec=rep(0, 2),
+             Amat=matrix(0, nr=2, nc=1),
+             bvec=0)
> # Minimum variance weights sum equal to 1
> optiml <- solve.QP(Dmat=2*covmat,
+             dvec=rep(0, 2),
+             Amat=matrix(1, nr=2, nc=1),
+             bvec=1)
> # Optimal value of objective function
> t(optiml$solution) %*% covmat %*% optiml$solution
> ## Perform simple optimization for reference
> # Objective function for simple optimization
> objfun <- function(x) {
+   x <- c(x, 1-x)
+   t(x) %*% covmat %*% x
+ }  ##  end objfun
> unlist(optimize(f=objfun, interval=c(-1, 2)))
```

Portfolio Optimization Using Package *quadprog*

The objective function is designed to minimize portfolio variance and maximize its returns:

$$O(x) = \mathbf{w}^T \mathbb{C} \, \mathbf{w} - \mathbf{w}^T \mathbf{r}$$

Where $\mathbb{C}$ is the covariance matrix of returns, $\mathbf{r}$ is the vector of returns, and $\mathbf{w}$ is the vector of portfolio weights.

The portfolio weights $\mathbf{w}$ are constrained as:

$$\mathbf{w}^T \mathbb{1} = \sum_{i=1}^{n} w_i = 1$$

$$0 \leq w_i \leq 1$$

The function `solve.QP()` has the arguments:

`Dmat` and `dvec` are the matrix and vector defining the quadratic objective function.

`Amat` and `bvec` are the matrix and vector defining the constraints.

`meq` specifies the number of equality constraints (the first `meq` constraints are equalities, and the rest are inequalities).

```
> # Calculate daily ETF percentage returns
> symbolv <- c("VTI", "IEF", "DBC")
> nstocks <- NROW(symbolv)
> retp <- na.omit(rutils::etfenv$returns[, symbolv])
> # Calculate the covariance matrix
> covmat <- cov(retp)
> # Minimum variance weights, with sum equal to 1
> optiml <- quadprog::solve.QP(Dmat=2*covmat,
+             dvec=numeric(3),
+             Amat=matrix(1, nr=3, nc=1),
+             bvec=1)
> # Minimum variance, maximum returns
> optiml <- quadprog::solve.QP(Dmat=2*covmat,
+             dvec=apply(0.1*retp, 2, mean),
+             Amat=matrix(1, nr=3, nc=1),
+             bvec=1)
> # Minimum variance positive weights, sum equal to 1
> a_mat <- cbind(matrix(1, nr=3, nc=1),
+        diag(3), -diag(3))
> b_vec <- c(1, rep(0, 3), rep(-1, 3))
> optiml <- quadprog::solve.QP(Dmat=2*covmat,
+             dvec=numeric(3),
+             Amat=a_mat,
+             bvec=b_vec,
+             meq=1)
```

The Covariance of Stock Returns

Estimating the covariance of stock returns is complicated because their date ranges may not overlap in time. Stocks may trade over different date ranges because of IPOs and corporate events (takeovers, mergers).

The function `cov()` calculates the covariance matrix of time series. The argument `use="pairwise.complete.obs"` removes `NA` values from pairs of stock returns.

But removing `NA` values in pairs of stock returns can produce covariance matrices which are not positive semi-definite.

The reason is because the covariance are calculated over different time intervals for different pairs of stock returns.

Matrices which are not positive semi-definite may not have an inverse matrix, but they have a generalized inverse.

The function `MASS::ginv()` calculates the generalized inverse of a matrix.

```
> # Select all the ETF symbols except "VXX", "SVXY" "MTUM", "QUAL",
> symbolv <- colnames(rutils::etfenv$returns)
> symbolv <- symbolv[!(symbolv %in% c("VXX", "SVXY", "MTUM", "QUAL",
> # Extract columns of rutils::etfenv$returns and overwrite NA value
> retp <- rutils::etfenv$returns["1999/", symbolv]
> sum(is.na(retp)) ##  Number of NA values
> nstocks <- NCOL(retp)
> datev <- zoo::index(retp)
> # Calculate the covariance ignoring NA values
> covmat <- cov(retp, use="pairwise.complete.obs")
> sum(is.na(covmat))
> # Calculate the inverse of covmat
> invmat <- solve(covmat)
> round(invmat %*% covmat, digits=5)
> # Calculate the generalized inverse of covmat
> invreg <- MASS::ginv(covmat)
> all.equal(unname(invmat), invreg)
```

Generalized Inverse of Singular Covariance Matrices

The standard inverse of a positive semi-definite matrix $\mathbb{C}$ can be calculated from its *eigenvalues* $\mathbb{D}$ and its *eigenvectors* $\mathbb{O}$ as follows:

$$\mathbb{C}^{-1} = \mathbb{O}\,\mathbb{D}^{-1}\,\mathbb{O}^T$$

The covariance matrix may not be positive semi-definite if the number of time periods of returns (rows) is less than the number of stocks (columns).

In that case some of the higher order eigenvalues are zero, and the above covariance matrix inverse is singular.

But a non-positive semi-definite covariance matrix may still have a *generalized inverse*.

The *generalized inverse* $\mathbb{C}_g^{-1}$ is calculated by removing the zero eigenvalues, and keeping only the first n non-zero *eigenvalues*:

$$\mathbb{C}_g^{-1} = \mathbb{O}_n\,\mathbb{D}_n^{-1}\,\mathbb{O}_n^T$$

Where $\mathbb{D}_n$ and $\mathbb{O}_n$ are matrices with the higher order eigenvalues and eigenvectors removed.

The generalized inverse $\mathbb{C}_g^{-1}$ of the matrix $\mathbb{C}$ satisfies the equation:

$$\mathbb{C}\,\mathbb{C}_g^{-1}\mathbb{C} = \mathbb{C}$$

Which is a generalization of the standard inverse property: $\mathbb{C}^{-1}\mathbb{C} = \mathbb{1}$

```
> # Create rectangular matrix with collinear columns
> matv <- matrix(rnorm(10*8), nc=10)
> # Calculate covariance matrix
> covmat <- cov(matv)
> # Calculate inverse of covmat - error
> invmat <- solve(covmat)
> # Perform eigen decomposition
> eigend <- eigen(covmat)
> eigenvec <- eigend$vectors
> eigenval <- eigend$values
> # Set tolerance for determining zero singular values
> precv <- sqrt(.Machine$double.eps)
> # Calculate generalized inverse from the eigen decomposition
> notzero <- (eigenval > (precv*eigenval[1]))
> inveigen <- eigenvec[, notzero] %*%
+   (t(eigenvec[, notzero]) / eigenval[notzero])
> # Inverse property of inveigen isn't satisfied
> all.equal(inveigen %*% covmat, diag(NROW(covmat)))
> # Generalized inverse property of inveigen is satisfied
> all.equal(covmat %*% inveigen %*% covmat, covmat)
> # Calculate generalized inverse using MASS::ginv()
> invreg <- MASS::ginv(covmat)
> # Verify that inveigen is the same as invreg
> all.equal(inveigen, invreg)
```

Portfolio Optimization Strategy

The optimal portfolio weights $\mathbf{w}$ are equal to the past in-sample excess returns $\mu = \mathbf{r} - r_f$ (in excess of the risk-free rate r_f) multiplied by the inverse $\mathbb{C}^{-1}$ of the covariance matrix $\mathbb{C}$:

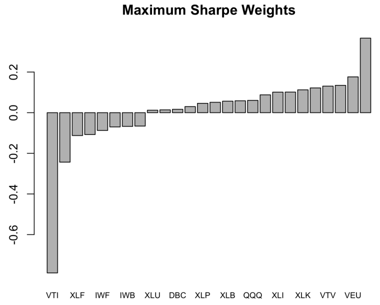
$$\mathbf{w} = \mathbb{C}^{-1} \mu$$

The *portfolio optimization* strategy invests in the best performing portfolio in the past *in-sample* interval, expecting that it will continue performing well *out-of-sample*.

The *portfolio optimization* strategy consists of:

1. Calculating the maximum Sharpe ratio portfolio weights in the *in-sample* interval,
2. Applying the weights and calculating the portfolio returns in the *out-of-sample* interval.

```
> # Maximum Sharpe weights in-sample interval
> retis <- retp["/2014"]
> raterf <- 0.03/252
> retx <- (retis - raterf)
> invreg <- MASS::ginv(cov(retis, use="pairwise.complete.obs"))
> weightv <- drop(invreg %*% colMeans(retx, na.rm=TRUE))
> weightv <- weightv/sqrt(sum(weightv^2))
> names(weightv) <- colnames(retp)
```



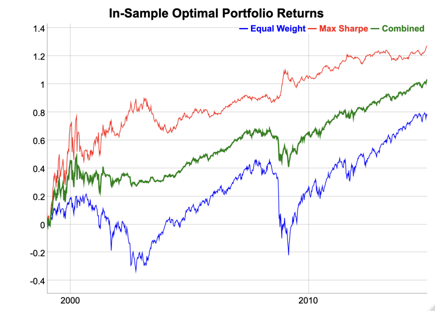

```
> # Plot portfolio weights
> barplot(sort(weightv), main="Maximum Sharpe Weights", cex.names=0.
```

Portfolio Optimization Strategy In-Sample

The in-sample performance of the *maximum Sharpe* portfolio is much better than the equal weight portfolio.

The function HighFreq::mult_mat() multiplies element-wise the rows or columns of a matrix times a vector.

```
> # Calculate the equal weight index
> retew <- xts::xts(rowMeans(retp, na.rm=TRUE), datev)
> # Calculate the in-sample weighted returns using Rcpp
> pnlis <- HighFreq::mult_mat(weightv, retis)
> # Or using the transpose
> # pnlis <- unname(t(t(retis)*weightv))
> pnlis <- rowMeans(pnlis, na.rm=TRUE)
> pnlis <- pnlis*sd(retew["/2014"])/sd(pnlis)
```



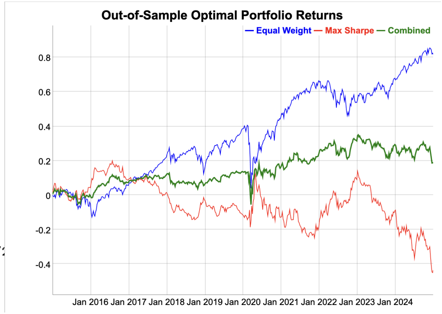

```
> # Calculate the Sharpe and Sortino ratios
> wealthv <- cbind(retew["/2014"], pnlis, (pnlis + retew["/2014"])/2
> colnames(wealthv) <- c("EqualWeight", "Max Sharpe", "Combined")
> sqrt(252)*sapply(wealthv, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Dygraph cumulative wealth
> endw <- rutils::calc_endpoints(wealthv, interval="weeks")
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="In-Sample Optimal Portfolio Returns") %>%
+   dyOptions(colors=c("blue", "red", "green"), strokeWidth=1) %>%
+   dySeries(name="Combined", strokeWidth=2) %>%
+   dyLegend(width=300)
```

Portfolio Optimization Strategy Out-of-Sample

The out-of-sample performance of the optimal in-sample portfolio is not nearly as good as in-sample.

Combining the *maximum Sharpe* portfolio with the equal weight portfolio produces and even better performing portfolio.

```
> # Calculate the equal weight index
> retos <- retp["2015/"]
> # Calculate out-of-sample portfolio returns
> pnlos <- HighFreq::mult_mat(weightv, retos)
> pnlos <- rowMeans(pnlos, na.rm=TRUE)
> pnlos <- pnlos*sd(retew["2015/"])/sd(pnlos)
> wealthv <- cbind(retew["2015/"], pnlos, (pnlos + retew["2015/"])/2
> colnames(wealthv) <- c("EqualWeight", "Max Sharpe", "Combined")
> # Calculate the Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
```



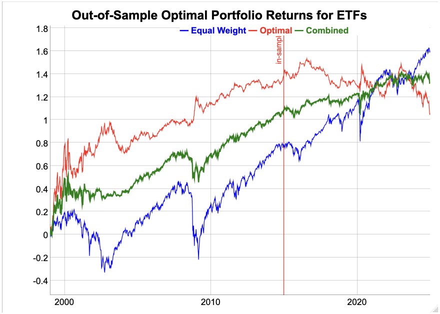

```
> # Dygraph cumulative wealth
> endw <- rutils::calc_endpoints(wealthv, interval="weeks")
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Out-of-Sample Optimal Portfolio Returns") %>%
+   dyOptions(colors=c("blue", "red", "green"), strokeWidth=1) %>%
+   dySeries(name="Combined", strokeWidth=2) %>%
+   dyLegend(width=300)
```

Portfolio Optimization Strategy for ETFs

The *portfolio optimization* strategy for ETFs is *overfit* in the *in-sample* interval.

Therefore the strategy doesn't perform as well in the *out-of-sample* interval as in the *in-sample* interval.

```
> # Maximum Sharpe weights in-sample interval
> covis <- cov(retis, use="pairwise.complete.obs")
> invreg <- MASS::ginv(covis)
> weightv <- invreg %*% colMeans(retx, na.rm=TRUE)
> names(weightv) <- colnames(retp)
> # Calculate cumulative wealth
> pnls <- HighFreq::mult_mat(weightv, retp)
> pnls <- rowMeans(pnls, na.rm=TRUE)
> pnls <- pnls*sd(retew)/sd(pnls)
> wealthv <- cbind(retew, pnls, (pnls + retew)/2)
> colnames(wealthv) <- c("EqualWeight", "Optimal", "Combined")
> # Calculate the in-sample Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv["/2014"], function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Calculate the Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
```




```
> # Dygraph cumulative wealth
> endw <- rutils::calc_endpoints(wealthv, interval="weeks")
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Out-of-Sample Optimal Portfolio Returns for ETFs") %>%
+   dyOptions(colors=c("blue", "red", "green"), strokeWidth=1) %>%
+   dySeries(name="Combined", strokeWidth=2) %>%
+   dyEvent(zoo::index(last(retis[, 1])), label="in-sample", strokeP
+   dyLegend(width=400)
```

Dimension Reduction of the Covariance Matrix

If the higher order singular values are very small then the inverse matrix amplifies the statistical noise in the response matrix.

The *reduced inverse* $\mathbb{C}_R^{-1}$ is calculated from the largest (lowest order) eigenvalues, up to *dimax* $= d$:

$$\mathbb{C}_R^{-1} = \mathbb{O}_d \, \mathbb{D}_d^{-1} \, \mathbb{O}_d^T$$

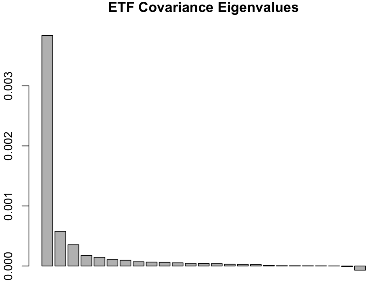
The parameter *dimax* specifies the number of eigenvalues used for calculating the *reduced inverse* of the covariance matrix of returns.

The *dimension reduction* technique calculates the *reduced inverse* of a covariance matrix by removing the very small, higher order eigenvalues, to reduce the propagation of statistical noise and improve the signal-to-noise ratio:

Even though the *reduced inverse* $\mathbb{C}_R^{-1}$ does not satisfy the matrix inverse property (so it's biased), its out-of-sample forecasts are usually more accurate than those using the exact inverse matrix.

But removing a larger number of eigenvalues increases the bias of the covariance matrix, which is an example of the *bias-variance tradeoff*.

The optimal value of the parameter *dimax* can be determined using *backtesting* (*cross-validation*).

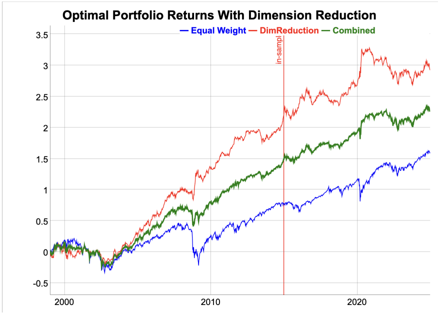



```
> # Calculate the eigen decomposition of the in-sample covariance ma
> eigend <- eigen(covis)
> eigenvec <- eigend$vectors
> eigenval <- eigend$values
> # Plot the eigenvalues
> barplot(eigenval, main="ETF Covariance Eigenvalues", cex.names=0.7
> # Calculate reduced inverse of covariance matrix
> dimax <- 9
> invred <- eigenvec[, 1:dimax] %*%
+   (t(eigenvec[, 1:dimax]) / eigenval[1:dimax])
> # Reduced inverse does not satisfy matrix inverse property
> all.equal(covis %*% invred %*% covis, covis)
```

Portfolio Optimization for ETFs with Dimension Reduction

The *out-of-sample* performance of the *portfolio optimization* strategy can be improved by applying dimension reduction to the inverse of the covariance matrix.

For small portfolios, the *in-sample* Sharpe ratio may also improve because dimension reduction reduces the portfolio volatility.




```
> # Calculate portfolio weights
> weightv <- invred %*% colMeans(retx, na.rm=TRUE)
> names(weightv) <- colnames(retp)
> # Calculate cumulative wealth
> pnls <- HighFreq::mult_mat(weightv, retp)
> pnls <- rowMeans(pnls, na.rm=TRUE)
> pnls <- pnls*sd(retew)/sd(pnls)
> wealthv <- cbind(retew, pnls, (pnls + retew)/2)
> colnames(wealthv) <- c("EqualWeight", "DimReduction", "Combined")
> # Calculate the in-sample Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv["/2014"], function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Calculate the Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
```

```
> # Dygraph cumulative wealth
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Optimal Portfolio Returns With Dimension Reduction") %>%
+   dyOptions(colors=c("blue", "red", "green"), strokeWidth=1) %>%
+   dySeries(name="Combined", strokeWidth=2) %>%
+   dyEvent(zoo::index(last(retis[, 1])), label="in-sample", strokeP
+   dyLegend(width=400)
```

Portfolio Optimization With Return Shrinkage

To further reduce the statistical noise, the individual returns r_i can be *shrunk* to the average portfolio returns $\bar{r}$:

$$r_i' = (1 - \alpha)\, r_i + \alpha\, \bar{r}$$

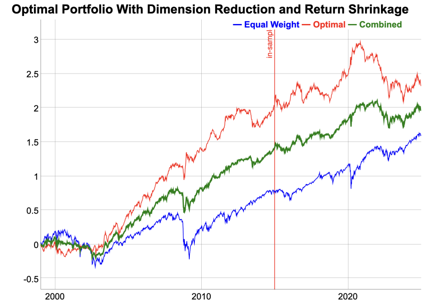
The parameter α is the *shrinkage* intensity, and it determines the strength of the *shrinkage* of individual returns to their mean.

If $\alpha = 0$ then there is no *shrinkage*, while if $\alpha = 1$ then all the returns are *shrunk* to their common mean: $r_i = \bar{r}$.

If $\alpha = 1$ then the weights are equal to the minimum variance portfolio weights. So an intermediate value of α gives weights that are an average of the maximum Sharpe weights and the minimum variance weights.

The optimal value of the *shrinkage* intensity α can be determined using *backtesting* (*cross-validation*).

```
> # Shrink the in-sample returns to their mean
> alphac <- 0.7
> retxm <- rowMeans(retx, na.rm=TRUE)
> retxis <- (1-alphac)*retx + alphac*retxm
> # Calculate the portfolio weights
> weightv <- invred %*% colMeans(retxis, na.rm=TRUE)
> # Calculate the cumulative wealth
> pnls <- HighFreq::mult_mat(weightv, retp)
> pnls <- rowMeans(pnls, na.rm=TRUE)
> pnls <- pnls*sd(retew)/sd(pnls)
> wealthv <- cbind(retew, pnls, (pnls + retew)/2)
> colnames(wealthv) <- c("EqualWeight", "Optimal", "Combined")
```




```
> # Calculate the Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Dygraph cumulative wealth
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Optimal Portfolio With Dimension Reduction and Return Shri
+   dyOptions(colors=c("blue", "red", "green"), strokeWidth=1) %>%
+   dySeries(name="Combined", strokeWidth=2) %>%
+   dyEvent(zoo::index(last(retis[, 1])), label="in-sample", strokeP
+   dyLegend(width=300)
```

Portfolio Momentum Strategy

In a *portfolio momentum strategy*, the portfolio is optimized periodically and held out-of-sample.

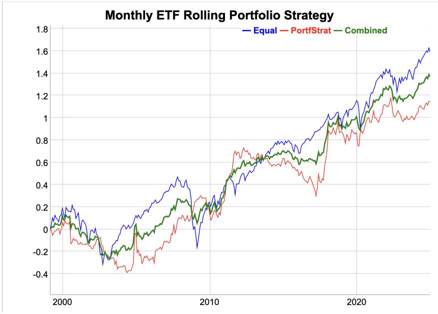
- Calculate the *end points* for portfolio rebalancing,
- Define an objective function for optimizing the portfolio weights,
- Calculate the optimal portfolio weights from the past (in-sample) performance,
- Calculate the out-of-sample returns by applying the portfolio weights to the future returns.

```
> # Define monthly end and start points
> retx <- (retp - raterf)
> endd <- rutils::calc_endpoints(retp, interval="months")
> endd <- endd[endd > (nstocks+1)]
> npts <- NROW(endd)
> lookb <- 3
> startp <- c(rep_len(0, lookb), endd[1:(npts-lookb)])
> # Perform loop over end points
> pnls <- lapply(1:(npts-1), function(tday) {
+     ##  Calculate the portfolio weights
+     retis <- retx[startp[tday]:endd[tday], ]
+     covmat <- cov(retis, use="pairwise.complete.obs")
+     covmat[is.na(covmat)] <- 0
+     invreg <- MASS::ginv(covmat)
+     colm <- colMeans(retis, na.rm=TRUE)
+     colm[is.na(colm)] <- 0
+     weightv <- invreg %*% colm
+     ##  Calculate the in-sample portfolio returns
+     pnlis <- HighFreq::mult_mat(weightv, retis)
+     pnlis <- rowMeans(pnlis, na.rm=TRUE)
+     ##  Calculate the out-of-sample portfolio returns
+     retos <- retp[(endd[tday]+1):endd[tday+1], ]
+     pnlos <- HighFreq::mult_mat(weightv, retos)
+     pnlos <- rowMeans(pnlos, na.rm=TRUE)
+     xts::xts(pnlos, zoo::index(retos))
+ })  ##  end lapply
> pnls <- do.call(rbind, pnls)
> pnls <- rbind(retew[paste0("/", start(pnls)-1)], pnls)
```

Portfolio Momentum Strategy Performance

In a *portfolio momentum strategy*, the portfolio is optimized periodically and held out-of-sample.

```
> # Calculate the Sharpe and Sortino ratios
> pnls <- pnls*sd(retew)/sd(pnls)
> wealthv <- cbind(retew, pnls, (pnls + retew)/2)
> colnames(wealthv) <- c("EqualWeight", "PortfStrat", "Combined")
> sqrt(252)*sapply(wealthv, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
```



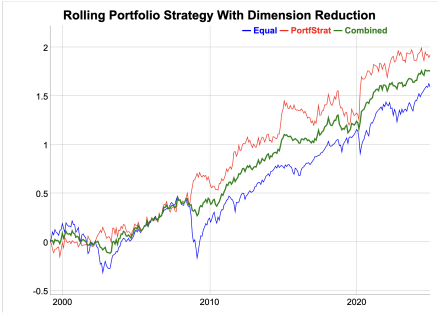

```
> # Dygraph cumulative wealth
> dygraphs::dygraph(cumsum(wealthv)[endw], main="Monthly ETF Portfol
+   dyOptions(colors=c("blue", "red", "green"), strokeWidth=1) %>%
+   dySeries(name="Combined", strokeWidth=2) %>%
+   dyLegend(show="always", width=300)
```

Portfolio Momentum Strategy With Dimension Reduction

Dimension reduction improves the performance of the portfolio momentum strategy because it suppresses the data noise.

The strategy performed especially well during sharp market selloffs, like in the years 2008 and 2020.

```
> # Perform loop over end points
> dimax <- 9
> pnls <- lapply(1:(npts-1), function(tday) {
+     ##  Calculate the portfolio weights
+     retis <- retx[startp[tday]:endd[tday], ]
+     covmat <- cov(retis, use="pairwise.complete.obs")
+     covmat[is.na(covmat)] <- 0
+     eigend <- eigen(covmat)
+     eigenvec <- eigend$vectors
+     eigenval <- eigend$values
+     invred <- eigenvec[, 1:dimax] %*%
+ (t(eigenvec[, 1:dimax]) / eigenval[1:dimax])
+     colm <- colMeans(retis, na.rm=TRUE)
+     colm[is.na(colm)] <- 0
+     weightv <- invred %*% colm
+     ##  Calculate the in-sample portfolio returns
+     pnlis <- HighFreq::mult_mat(weightv, retis)
+     pnlis <- rowMeans(pnlis, na.rm=TRUE)
+     weightv <- weightv*0.01/sd(pnlis)
+     ##  Calculate the out-of-sample portfolio returns
+     retos <- retp[(endd[tday]+1):endd[tday+1], ]
+     pnlos <- HighFreq::mult_mat(weightv, retos)
+     pnlos <- rowMeans(pnlos, na.rm=TRUE)
+     xts::xts(pnlos, zoo::index(retos))
+ })  ##  end lapply
> pnls <- do.call(rbind, pnls)
> pnls <- rbind(retew[paste0("/", start(pnls)-1)], pnls)
```



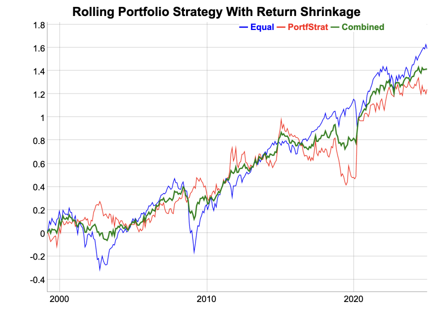

```
> # Calculate the Sharpe and Sortino ratios
> pnls <- pnls*sd(retew)/sd(pnls)
> wealthv <- cbind(retew, pnls, (pnls + retew)/2)
> colnames(wealthv) <- c("EqualWeight", "PortfStrat", "Combined")
> sqrt(252)*sapply(wealthv, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Dygraph cumulative wealth
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Portfolio Momentum Strategy With Dimension Reduction") %>%
+   dyOptions(colors=c("blue", "red", "green"), strokeWidth=1) %>%
+   dySeries(name="Combined", strokeWidth=2) %>%
+   dyLegend(show="always", width=300)
```

Portfolio Momentum Strategy With Return Shrinkage

Return shrinkage averages the stock returns, which creates bias, but it also reduces the variance.

Return shrinkage can be applied to improve the performance of the portfolio momentum strategy.

```
> alphac <- 0.7 ##  Return shrinkage intensity
> # Perform loop over end points
> pnls <- lapply(1:(npts-1), function(tday) {
+     ##  Shrink the in-sample returns to their mean
+     retis <- retx[startp[tday]:endd[tday], ]
+     rowm <- rowMeans(retis, na.rm=TRUE)
+     rowm[is.na(rowm)] <- 0
+     retis <- (1-alphac)*retis + alphac*rowm
+     ##  Calculate the portfolio weights
+     covmat <- cov(retis, use="pairwise.complete.obs")
+     covmat[is.na(covmat)] <- 0
+     eigend <- eigen(covmat)
+     eigenvec <- eigend$vectors
+     eigenval <- eigend$values
+     invred <- eigenvec[, 1:dimax] %*%
+ (t(eigenvec[, 1:dimax]) / eigenval[1:dimax])
+     colm <- colMeans(retis, na.rm=TRUE)
+     colm[is.na(colm)] <- 0
+     weightv <- invred %*% colm
+     ##  Scale the weights to volatility target
+     pnlis <- HighFreq::mult_mat(weightv, retis)
+     pnlis <- rowMeans(pnlis, na.rm=TRUE)
+     weightv <- weightv*0.01/sd(pnlis)
+     ##  Calculate the out-of-sample portfolio returns
+     retos <- retp[(endd[tday]+1):endd[tday+1], ]
+     pnlos <- HighFreq::mult_mat(weightv, retos)
+     pnlos <- rowMeans(pnlos, na.rm=TRUE)
+     xts::xts(pnlos, zoo::index(retos))
+ })  ##  end lapply
> pnls <- do.call(rbind, pnls)
> pnls <- rbind(retew[paste0("/", start(pnls)-1)], pnls)
```




```
> # Calculate the Sharpe and Sortino ratios
> pnls <- pnls*sd(retew)/sd(pnls)
> wealthv <- cbind(retew, pnls, (pnls + retew)/2)
> colnames(wealthv) <- c("EqualWeight", "PortfStrat", "Combined")
> sqrt(252)*sapply(wealthv, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Dygraph cumulative wealth
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Portfolio Momentum Strategy With Return Shrinkage") %>%
+   dyOptions(colors=c("blue", "red", "green"), strokeWidth=1) %>%
+   dySeries(name="Combined", strokeWidth=2) %>%
+   dyLegend(show="always", width=300)
```

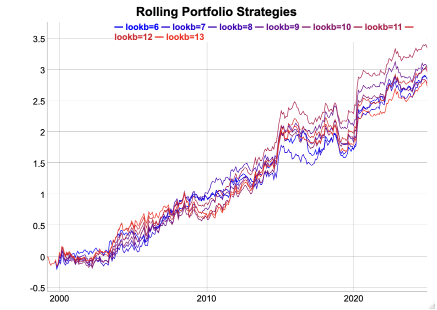
Function for Portfolio Momentum Strategy

```
> # Define backtest functional for portfolio momentum strategy
> roll_portf <- function(retx, ##  Excess returns
+                  retp, ##  Stock returns
+                  endd, ##  End points
+                  lookb=12, ##  Look-back interval
+                  dimax=3, ##  Dimension reduction parameter
+                  alphac=0.0, ##  Return shrinkage intensity
+                  bidask=0.0, ##  Bid-offer spread
+                  ...) {
+   npts <- NROW(endd)
+   startp <- c(rep_len(0, lookb), endd[1:(npts-lookb)])
+   pnls <- lapply(1:(npts-1), function(tday) {
+     retis <- retx[startp[tday]:endd[tday], ]
+     ##  Shrink the in-sample returns to their mean
+     if (alphac > 0) {
+ rowm <- rowMeans(retis, na.rm=TRUE)
+ rowm[is.na(rowm)] <- 0
+ retis <- (1-alphac)*retis + alphac*rowm
+     } ##  end if
+     ##  Calculate the portfolio weights
+     covmat <- cov(retis, use="pairwise.complete.obs")
+     covmat[is.na(covmat)] <- 0
+     eigend <- eigen(covmat)
+     eigenvec <- eigend$vectors
+     eigenval <- eigend$values
+     invred <- eigenvec[, 1:dimax] %*% (t(eigenvec[, 1:dimax]) / eigenval[1:dimax])
+     colm <- colMeans(retis, na.rm=TRUE)
+     colm[is.na(colm)] <- 0
+     weightv <- invred %*% colm
+     ##  Scale the weights to volatility target
+     pnlis <- HighFreq::mult_mat(weightv, retis)
+     pnlis <- rowMeans(pnlis, na.rm=TRUE)
+     weightv <- weightv*0.01/sd(pnlis)
+     ##  Calculate the out-of-sample portfolio returns
+     retos <- retp[(endd[tday]+1):endd[tday+1], ]
+     pnlos <- HighFreq::mult_mat(weightv, retos)
+     pnlos <- rowMeans(pnlos, na.rm=TRUE)
+     xts::xts(pnlos, zoo::index(retos))
```

Portfolio Momentum With Different Look-backs

Multiple *portfolio momentum* strategies can be backtested by calling the function roll_portf() in a loop over a vector of *look-back* parameters.

```
> # Simulate a monthly ETF portfolio strategy
> pnls <- roll_portf(retx=retx, retp=retp, endd=endd,
+   lookb=lookb, dimax=dimax)
> # Perform sapply loop over lookbv
> lookbv <- seq(6, 13, by=1)
> library(parallel)  ##  Load package parallel
> ncores <- detectCores() - 1
> pnls <- mclapply(lookbv, roll_portf, retp=retp, retx=retx,
+   endd=endd, dimax=dimax, mc.cores=ncores)
> pnls <- do.call(cbind, pnls)
> colnames(pnls) <- paste0("lookb=", lookbv)
> profilev <- sapply(pnls, sum)
> lookb <- lookbv[which.max(profilev)]
```

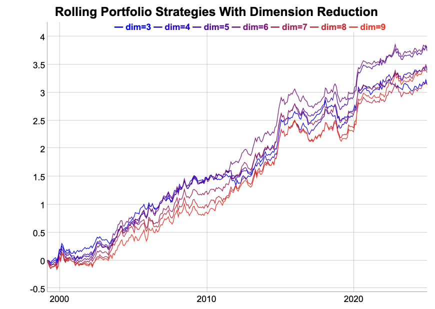



```
> # Calculate the Sharpe and Sortino ratios
> sqrt(252)*sapply(pnls, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Plot dygraph of monthly ETF portfolio strategies
> colorv <- colorRampPalette(c("blue", "red"))(NCOL(pnls))
> dygraphs::dygraph(cumsum(pnls)[endw], main="Portfolio Momentum Str
+   dyOptions(colors=colorv, strokeWidth=1) %>%
+   dyLegend(show="always", width=500)
> # Plot portfolio strategies using quantmod
> themev <- chart_theme()
> themev$col$line.col <-
+   colorRampPalette(c("blue", "red"))(NCOL(pnls))
> quantmod::chart_Series(cumsum(pnls),
+   theme=themev, name="Portfolio Momentum Strategies")
> legend("bottomleft", legend=colnames(pnls),
+   inset=0.02, bg="white", cex=0.7, lwd=rep(6, NCOL(retp)),
+   col=themev$col$line.col, bty="n")
```

Portfolio Momentum With Dimension Reduction

Multiple *portfolio momentum* strategies can be backtested by calling the function roll_portf() in a loop over a vector of the dimension reduction parameter.

```
> # Perform backtest for different dimax values
> dimv <- 3:9
> pnls <- mclapply(dimv, roll_portf, retp=retp, retx=retx,
+   endd=endd, lookb=lookb, mc.cores=ncores)
> pnls <- do.call(cbind, pnls)
> colnames(pnls) <- paste0("dim=", dimv)
> profilev <- sapply(pnls, sum)
> dimax <- dimv[which.max(profilev)]
```



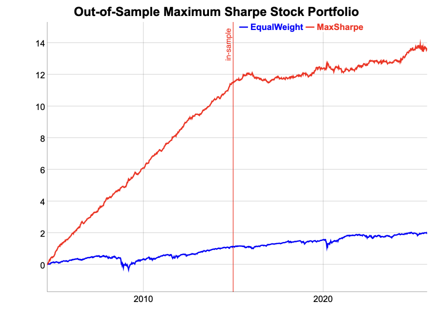

```
> # Calculate the Sharpe and Sortino ratios
> sqrt(252)*sapply(pnls, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Plot dygraph of monthly ETF portfolio strategies
> colorv <- colorRampPalette(c("blue", "red"))(NCOL(pnls))
> dygraphs::dygraph(cumsum(pnls)[endw],
+   main="Portfolio Momentum Strategies With Dimension Reduction") %
+   dyOptions(colors=colorv, strokeWidth=1) %>%
+   dyLegend(show="always", width=500)
> # Plot portfolio strategies using quantmod
> themev <- chart_theme()
> themev$col$line.col <-
+   colorRampPalette(c("blue", "red"))(NCOL(pnls))
> quantmod::chart_Series(cumsum(pnls),
+   theme=themev, name="Portfolio Momentum Strategies")
> legend("bottomleft", legend=colnames(pnls),
+   inset=0.02, bg="white", cex=0.7, lwd=rep(6, NCOL(retp)),
+   col=themev$col$line.col, bty="n")
```

Optimal Stock Portfolio Out-of-Sample

The *portfolio optimization* strategy for stocks is *overfit* in the *in-sample* interval.

Therefore the strategy performance is poor in the *out-of-sample* interval.

```
> # Load stock returns
> load("/Users/jerzy/Develop/lecture_slides/data/sp500_returns.RData
> datev <- zoo::index(na.omit(retstock$GOOGL))
> retp <- retstock[datev] ##  Subset the returns to GOOGL
> # Remove the stocks with any NA values
> numna <- sapply(retp, function(x) sum(is.na(x)))
> retp <- retp[, numna==0]
> nstocks <- NCOL(retp)
> datev <- zoo::index(retp)
> symbolv <- colnames(retp)
> retis <- retp["/2014"] ##  In-sample returns
> raterf <- 0.03/252
> retx <- (retis - raterf) ##  Excess returns
> # Maximum Sharpe weights in-sample interval
> covmat <- cov(retis, use="pairwise.complete.obs")
> invreg <- MASS::ginv(covmat)
> colmeanv <- colMeans(retx, na.rm=TRUE)
> weightv <- drop(invreg %*% colmeanv)
> names(weightv) <- symbolv
> head(sort(weightv))
> tail(sort(weightv))
> # Calculate the portfolio returns
> pnls <- HighFreq::mult_mat(weightv, retp)
> pnls <- rowMeans(pnls, na.rm=TRUE)
> pnls <- xts::xts(pnls, datev)
> retew <- xts::xts(rowMeans(retp, na.rm=TRUE), datev)
> pnls <- pnls*sd(retew["/2014"])/sd(pnls["/2014"])
> wealthv <- cbind(retew, pnls)
> colnames(wealthv) <- c("EqualWeight", "MaxSharpe")
```



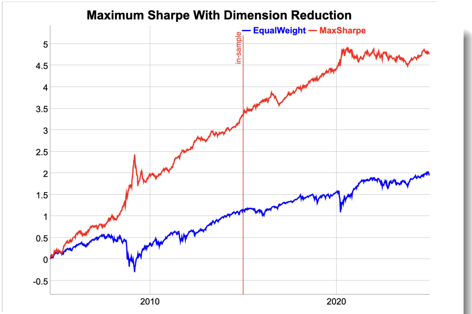

```
> # Calculate the in-sample Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv["/2014"], function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Calculate the out-of-sample Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv["2015/"], function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Plot of cumulative portfolio returns
> endw <- rutils::calc_endpoints(wealthv, interval="weeks")
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Out-of-Sample Maximum Sharpe Stock Portfolio") %>%
+   dyOptions(colors=c("blue", "red"), strokeWidth=2) %>%
+   dyEvent(zoo::index(last(retis[, 1])), label="in-sample", strokeP
+   dyLegend(width=300)
```

Maximum Sharpe Portfolio With Dimension Reduction

The *out-of-sample* performance of the *portfolio optimization* strategy can be improved by applying dimension reduction to the inverse of the covariance matrix.

The *in-sample* performance is worse because dimension reduction reduces *overfitting*.

```
> # Calculate reduced inverse of covariance matrix
> dimax <- 10
> eigend <- eigen(covmat)
> eigenvec <- eigend$vectors
> eigenval <- eigend$values
> invred <- eigenvec[, 1:dimax] %*%
+   (t(eigenvec[, 1:dimax]) / eigenval[1:dimax])
> # Calculate portfolio weights and PnLs
> colmeanv <- colMeans(retx["/2014"], na.rm=TRUE)
> weightv <- invred %*% colmeanv
> pnls <- HighFreq::mult_mat(weightv, retp)
> pnls <- rowMeans(pnls, na.rm=TRUE)
> pnls <- xts::xts(pnls, datev)
> pnls <- pnls*sd(retew["/2014"])/sd(pnls["/2014"])
```




```
> # Combine with equal weight
> wealthv <- cbind(retew, pnls)
> colnames(wealthv) <- c("EqualWeight", "MaxSharpe")
> # Calculate the in-sample Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv["/2014"], function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Calculate the out-of-sample Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv["2015/"], function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Plot of cumulative portfolio returns
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Maximum Sharpe With Dimension Reduction") %>%
+   dyOptions(colors=c("blue", "red"), strokeWidth=2) %>%
+   dyEvent(zoo::index(last(retis[, 1])), label="in-sample", strokeP
+   dyLegend(width=300)
```

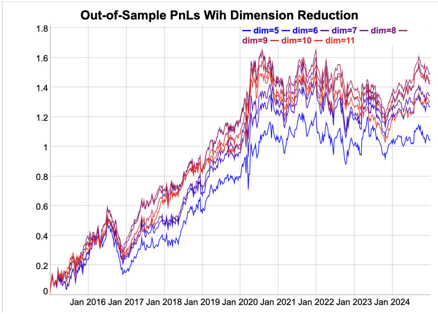
Optimal Dimension Reduction Out-of-Sample

The optimal out-of-sample dimension reduction parameter is `dimax=9`, which represents relatively weak dimension reduction.

The dependence of the out-of-sample strategy pnls on the *dimax* parameter reflects the *bias-variance tradeoff*.

If the *dimax* parameter is too small, it creates excessive bias, but if it's too large, it doesn't suppress the variance enough.

```
> # Perform loop over vector of dimension reduction parameters
> dimv <- seq(from=5, to=11)
> pnls <- mclapply(dimv, function(dimax) {
+   invred <- eigenvec[, 1:dimax] %*%
+     (t(eigenvec[, 1:dimax]) / eigenval[1:dimax])
+   ##  Calculate portfolio weights and PnLs
+   weightv <- invred %*% colmeanv
+   pnls <- HighFreq::mult_mat(weightv, retp)
+   pnls <- rowMeans(pnls, na.rm=TRUE)
+   pnls <- xts::xts(pnls, datev)
+   pnls*sd(retew["/2014"])/sd(pnls["/2014"])
+ }, mc.cores=ncores)  ##  end mclapply
> pnls <- do.call(cbind, pnls)
> colnames(pnls) <- paste0("dim=", dimv)
> profilev <- sapply(pnls["2015/"], sum)
> dimax <- dimv[which.max(profilev)]
```




```
> # Plot of cumulative portfolio returns
> colorv <- colorRampPalette(c("blue", "red"))(NCOL(pnls))
> endw <- rutils::calc_endpoints(pnls["2015/"], interval="weeks")
> dygraphs::dygraph(cumsum(pnls["2015/"])[endw],
+   main="Out-of-Sample PnLs Wih Dimension Reduction") %>%
+   dyOptions(colors=colorv, strokeWidth=1) %>%
+   dyLegend(show="always", width=300)
```

Maximum Sharpe Portfolio With Return Shrinkage

To further reduce the statistical noise, the individual returns r_i can be *shrunk* to the average portfolio returns $\bar{r}$:

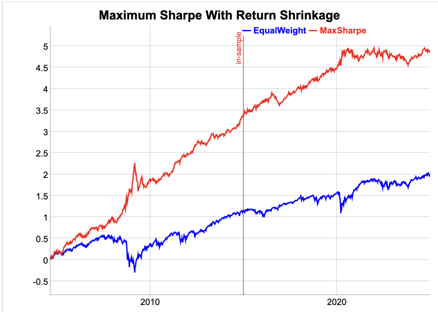
$$r'_i = (1 - \alpha)\, r_i + \alpha\, \bar{r}$$

The parameter α is the *shrinkage* intensity, and it determines the strength of the *shrinkage* of individual returns to their mean.

If $\alpha = 0$ then there is no *shrinkage*, while if $\alpha = 1$ then all the returns are *shrunk* to their common mean: $r_i = \bar{r}$.

The optimal value of the *shrinkage* intensity α can be determined using *backtesting* (*cross-validation*).

```
> # Shrink the in-sample returns to their mean
> alphac <- 0.3
> retxm <- rowMeans(retx, na.rm=TRUE)
> retxis <- (1-alphac)*retx + alphac*retxm
> # Calculate portfolio weights and PnLs
> weightv <- drop(invred %*% colMeans(retxis, na.rm=TRUE))
> names(weightv) <- symbolv
> pnls <- HighFreq::mult_mat(weightv, retp)
> pnls <- rowMeans(pnls, na.rm=TRUE)
> pnls <- xts::xts(pnls, datev)
> pnls <- pnls*sd(retew["/2014"])/sd(pnls["/2014"])
```




```
> # Combine with equal weight
> wealthv <- cbind(retew, pnls)
> colnames(wealthv) <- c("EqualWeight", "MaxSharpe")
> # Calculate the in-sample Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv["/2014"], function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Calculate the out-of-sample Sharpe and Sortino ratios
> sqrt(252)*sapply(wealthv["2015/"], function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Plot of cumulative portfolio returns
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Maximum Sharpe With Return Shrinkage") %>%
+   dyOptions(colors=c("blue", "red"), strokeWidth=2) %>%
+   dyEvent(zoo::index(last(retis[, 1])), label="in-sample", strokeP
+   dyLegend(width=300)
```

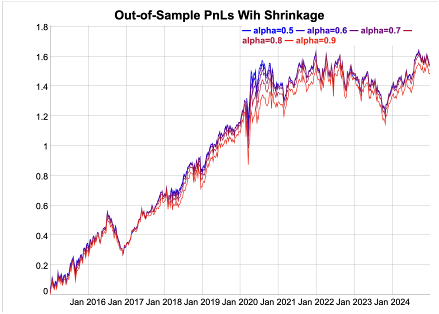
Optimal Shrinkage Parameter Out-of-Sample

The optimal out-of-sample return shrinkage parameter is equal to $\alpha = 0.6$, which represents moderate shrinkage.

The dependence of the out-of-sample strategy pnls on the α parameter reflects the *bias-variance tradeoff*.

If the α parameter is too large, it creates excessive bias, but if it's too small, it doesn't suppress the variance enough.

```
> # Perform loop over vector of shrinkage intensities
> alphav <- seq(from=0.5, to=0.9, by=0.1)
> pnls <- mclapply(alphav, function(alphac) {
+   retxis <- (1-alphac)*retx + alphac*retxm
+   weightv <- drop(invred %*% colMeans(retxis, na.rm=TRUE))
+   pnls <- HighFreq::mult_mat(weightv, retp)
+   pnls <- rowMeans(pnls, na.rm=TRUE)
+   pnls <- xts::xts(pnls, datev)
+   pnls*sd(retew["/2014"])/sd(pnls["/2014"])
+ }, mc.cores=ncores)  ##  end mclapply
> pnls <- do.call(cbind, pnls)
> colnames(pnls) <- paste0("alpha=", alphav)
> profilev <- sapply(pnls["2015/"], sum)
> alphac <- alphav[which.max(profilev)]
```




```
> # Plot of cumulative portfolio returns
> colorv <- colorRampPalette(c("blue", "red"))(NCOL(pnls))
> endw <- rutils::calc_endpoints(pnls["2015/"], interval="weeks")
> dygraphs::dygraph(cumsum(pnls["2015/"])[endw],
+   main="Out-of-Sample PnLs Wih Shrinkage") %>%
+   dyOptions(colors=colorv, strokeWidth=1) %>%
+   dyLegend(show="always", width=300)
```

Rolling Maximum Sharpe Stock Portfolio Strategy

A *portfolio momentum* strategy consists of rebalancing a portfolio over the end points:

1. Calculate the maximum Sharpe ratio portfolio weights at each end point,
2. Apply the weights in the next interval and calculate the out-of-sample portfolio returns.

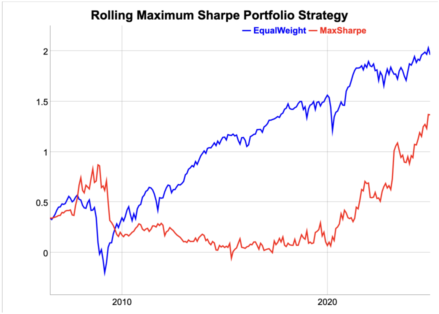
The strategy parameters are: the rebalancing frequency (annual, monthly, etc.), and the length of look-back interval.

```
> # Define monthly end points
> endd <- rutils::calc_endpoints(retp, interval="months")
> endd <- endd[endd > (nstocks+1)]
> npts <- NROW(endd) ; lookb <- 12
> startp <- c(rep_len(0, lookb), endd[1:(npts-lookb)])
> # !!! Perform parallel loop over end points - takes very long!!!
> library(parallel)  ##  Load package parallel
> ncores <- detectCores() - 1
> pnls <- mclapply(1:(npts-1), function(tday) {
+     ##  Subset the excess returns
+     retis <- retp[startp[tday]:endd[tday], ]
+     retis[is.na(retis)] <- 0
+     invreg <- MASS::ginv(cov(retis, use="pairwise.complete.obs"))
+     ##  Calculate the maximum Sharpe ratio portfolio weights
+     retm <- colMeans(retis, na.rm=TRUE)
+     weightv <- invreg %*% retm
+     ##  Zero weights if sparse data
+     zerov <- sapply(retis, function(x) (sum(x == 0) > 5))
+     weightv[zerov] <- 0
+     ##  Calculate in-sample portfolio returns
+     pnlis <- (retis %*% weightv)
+     ##  Scale the weights to the in-sample volatility
+     weightv <- weightv*sd(retm)/sd(pnlis)
+     ##  Calculate the out-of-sample portfolio returns
+     retos <- retp[(endd[tday]+1):endd[tday+1], ]
+     pnls <- HighFreq::mult_mat(weightv, retos)
+     pnls <- rowMeans(pnls, na.rm=TRUE)
+     xts::xts(pnls, zoo::index(retos))
+ }, mc.cores=ncores)  ##  end mclapply
> pnls <- rutils::do_call(rbind, pnls)
> pnls <- rbind(retew[paste0("/", start(pnls)-1)], pnls*sd(retew)/sc
```

Performance of Rolling Maximum Sharpe Strategy

The performance of the *portfolio momentum* strategy can be improved by applying dimension reduction and return shrinkage.

```
> # Calculate the Sharpe and Sortino ratios
> wealthv <- cbind(retew, pnls)
> colnames(wealthv) <- c("EqualWeight", "MaxSharpe")
> sqrt(252)*sapply(wealthv, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
```




```
> # Plot cumulative strategy returns
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Rolling Maximum Sharpe Portfolio Strategy") %>%
+   dyOptions(colors=c("blue", "red"), strokeWidth=2) %>%
+   dyLegend(show="always", width=300)
```

Fast Covariance Matrix Inverse Using *RcppArmadillo*

RcppArmadillo can be used to quickly calculate the reduced inverse of a covariance matrix.

```
> # Create random matrix of returns
> matv <- matrix(rnorm(300), nc=5)
> # Reduced inverse of covariance matrix
> dimax <- 3
> eigend <- eigen(covmat)
> invred <- eigend$vectors[, 1:dimax] %*%
+   (t(eigend$vectors[, 1:dimax]) / eigend$values[1:dimax])
> # Reduced inverse using RcppArmadillo
> invarma <- HighFreq::calc_inv(covmat, dimax)
> all.equal(invred, invarma)
> # Microbenchmark RcppArmadillo code
> library(microbenchmark)
> summary(microbenchmark(
+   rcode={eigend <- eigen(covmat)
+     eigend$vectors[, 1:dimax] %*%
+ (t(eigend$vectors[, 1:dimax]) / eigend$values[1:dimax])
+   },
+   rcpp=calc_inv(covmat, dimax),
+   times=10))[, c(1, 4, 5)]  ##  end microbenchmark summary
```

```
arma::mat calc_inv(const arma::mat& matv,
                   arma::uword dimax = 0, // Max number
                   double eigen_thresh = 0.01) { // Thre

  if (dimax == 0) {
    // Calculate the inverse using arma::pinv()
    return arma::pinv(tseries, eigen_thresh);
  } else {
    // Calculate the reduced inverse using SVD decomposi

    // Allocate SVD
    arma::vec svdval;
    arma::mat svdu, svdv;

    // Calculate the SVD
    arma::svd(svdu, svdval, svdv, tseries);

    // Subset the SVD
    dimax = dimax - 1;
    // For no regularization: dimax = tseries.n_cols
    svdu = svdu.cols(0, dimax);
    svdv = svdv.cols(0, dimax);
    svdval = svdval.subvec(0, dimax);

    // Calculate the inverse from the SVD
    return svdv*arma::diagmat(1/svdval)*svdu.t();

  }  // end if

}  // end calc_inv
```

Portfolio Optimization Using *RcppArmadillo*

Fast portfolio optimization using matrix algebra can be implemented using *RcppArmadillo*.

```
arma::vec calc_weights(const arma::mat& returns, // Asset returns
                       Rcpp::List controll) { // List of portfolio optimization parameters

  // Unpack the control list of portfolio optimization parameters
  // Type of portfolio optimization model
  std::string method = Rcpp::as<std::string>(controll["method"]);
  // Threshold level for discarding small singular values
  double eigen_thresh = Rcpp::as<double>(controll["eigen_thresh"]);
  // Dimension reduction
  arma::uword dimax = Rcpp::as<int>(controll["dimax"]);
  // Confidence level for calculating the quantiles of returns
  double confl = Rcpp::as<double>(controll["confl"]);
  // Shrinkage intensity of returns
  double alpha = Rcpp::as<double>(controll["alpha"]);
  // Should the weights be ranked?
  bool rankw = Rcpp::as<int>(controll["rankw"]);
  // Should the weights be centered?
  bool centerw = Rcpp::as<int>(controll["centerw"]);
  // Method for scaling the weights
  std::string scalew = Rcpp::as<std::string>(controll["scalew"]);
  // Volatility target for scaling the weights
  double vol_target = Rcpp::as<double>(controll["vol_target"]);

  // Initialize the variables
  arma::uword ncols = returns.n_cols;
  arma::vec weightv(ncols, fill::zeros);
  // If no regularization then set dimax to ncols
  if (dimax == 0)  dimax = ncols;
  // Calculate the covariance matrix
  arma::mat covmat = calc_covar(returns);

  // Apply different calculation methods for the weights
  switch(calc_method(method)) {
  case methodenum::maxsharpe: {
    // Mean returns of columns
```

Strategy Backtesting Using *RcppArmadillo*

Fast backtesting of strategies can be implemented using *RcppArmadillo*.

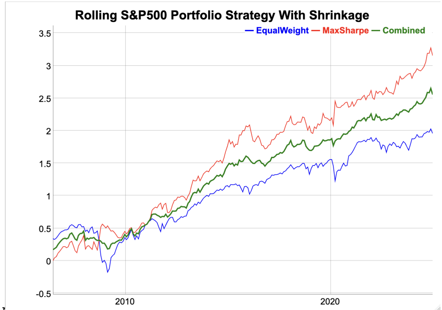
```
arma::mat roll_portf(const arma::mat& retx, // Asset excess returns
                    const arma::mat& retp, // Asset returns
                    Rcpp::List controll, // List of portfolio optimization model parameters
                    arma::uvec startp, // Start points
                    arma::uvec endp, // End points
                    double lambdaf = 0.0, // Decay factor for averaging the portfolio weights
                    double coeff = 1.0, // Multiplier of strategy returns
                    double bidask = 0.0) { // The bid-ask spread

  double lambda1 = 1-lambdaf;
  arma::uword nweights = retp.n_cols;
  arma::vec weightv(nweights, fill::zeros);
  arma::vec weights_past = arma::ones(nweights)/std::sqrt(nweights);
  arma::mat pnls = arma::zeros(retp.n_rows, 1);

  // Perform loop over the end points
  for (arma::uword it = 1; it < endp.size(); it++) {
    // cout << "it: " << it << endl;
    // Calculate the portfolio weights
    weightv = coeff*calc_weights(retx.rows(startp(it-1), endp(it-1)), controll);
    // Calculate the weights as the weighted sum with past weights
    weightv = lambda1*weightv + lambdaf*weights_past;
    // Calculate out-of-sample returns
    pnls.rows(endp(it-1)+1, endp(it)) = retp.rows(endp(it-1)+1, endp(it))*weightv;
    // Add transaction costs
    pnls.row(endp(it-1)+1) -= bidask*sum(abs(weightv - weights_past))/2;
    // Copy the weights
    weights_past = weightv;
  }  // end for

  // Return the strategy pnls
  return pnls;

}  // end roll_portf
```

Portfolio Momentum Strategy With Dimension Reduction and Shrinkage

The performance of the *portfolio momentum* strategy can be improved by applying dimension reduction and return shrinkage.

The *rolling portfolio* strategy has a higher *Sharpe* ratio and also higher absolute returns than the equal weight portfolio.

The backtest simulation can be performed very quickly using C++ code and package *RcppArmadillo*.




```
> # Shift the end points to C++ convention
> endd <- (endd - 1)
> endd[endd < 0] <- 0
> startp <- (startp - 1)
> startp[startp < 0] <- 0
> # Specify dimension reduction and return shrinkage using list of portfolio optimization parameters
> dimax <- 9
> alphac <- 0.8
> controll <- HighFreq::param_portf(method="maxsharpe",
+   dimax=dimax, alpha=alphac)
> # Perform backtest in Rcpp - takes very long!!!
> retx <- (retp - raterf)
> pnls <- HighFreq::roll_portf(retx=retx, retp=retp,
+   startp=startp, endd=endd, controll=controll)
> pnls <- pnls*sd(retew)/sd(pnls)
```

```
> # Calculate the out-of-sample Sharpe and Sortino ratios
> wealthv <- cbind(retew, pnls, (pnls + retew)/2)
> colnames(wealthv) <- c("EqualWeight", "MaxSharpe", "Combined")
> sqrt(252)*sapply(wealthv, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Plot cumulative strategy returns
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Rolling S&P500 Portfolio Strategy With Shrinkage") %>%
+   dyOptions(colors=c("blue", "red", "green"), strokeWidth=1) %>%
+   dySeries(name="Combined", label="Combined", strokeWidth=2) %>%
+   dyLegend(show="always", width=300)
```

Optimal Dimension Reduction Parameter

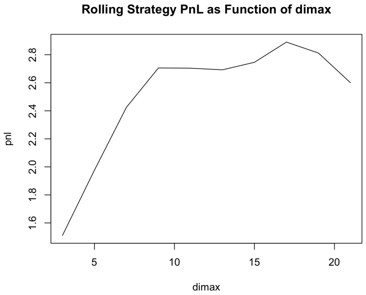
The optimal value of the dimension reduction parameter *dimax* can be determined using *backtesting*.

The optimal dimension reduction parameter for this portfolio of stocks is equal to `dimax=9`, which represents relatively weak dimension reduction.

The dependence of the out-of-sample strategy pnls on the *dimax* parameter reflects the *bias-variance tradeoff*.

If the *dimax* parameter is too small, it creates excessive bias, but if it's too large, it doesn't suppress the variance enough.

```
> # Perform backtest over vector of dimension reduction parameters
> dimv <- seq(from=3, to=21, by=2)
> pnls <- mclapply(dimv, function(dimax) {
+   controll <- HighFreq::param_portf(method="maxsharpe",
+     dimax=dimax, alpha=alphac)
+   HighFreq::roll_portf(retx=retx, retp=retp,
+     startp=startp, endd=endd, controll=controll)
+ }, mc.cores=ncores)  ##  end mclapply
> profilev <- sapply(pnls, sum)
> dimax <- dimv[which.max(profilev)]
```




```
> # Plot of rolling strategy PnL as a function of dimax
> plot(x=dimv, y=profilev, t="l", xlab="dimax", ylab="pnl",
+   main="Rolling Strategy PnL as Function of dimax")
```

Optimal Shrinkage Parameter

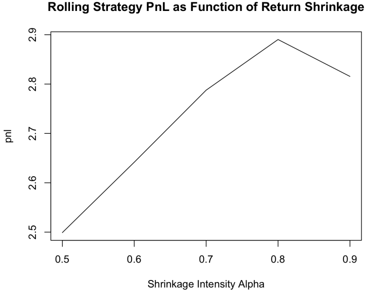
The optimal value of the return shrinkage intensity parameter α can be determined using *backtesting*.

The best return shrinkage parameter for this portfolio of stocks is equal to $\alpha = 0.8$, which represents strong shrinkage.

The dependence of the out-of-sample strategy pnls on the α parameter reflects the *bias-variance tradeoff*.

If the α parameter is too large, it creates excessive bias, but if it's too small, it doesn't suppress the variance enough.

```
> # Perform backtest over vector of shrinkage intensities
> alphav <- seq(from=0.5, to=0.9, by=0.1)
> pnls <- mclapply(alphav, function(alphac) {
+   controll <- HighFreq::param_portf(method="maxsharpe",
+       dimax=dimax, alpha=alphac)
+   HighFreq::roll_portf(retx=retx, retp=retp,
+       startp=startp, endd=endd, controll=controll)
+ }, mc.cores=ncores)  ##  end mclapply
> profilev <- sapply(pnls, sum)
> alphac <- alphav[which.max(profilev)]
```




```
> # Plot of rolling strategy PnL as a function of shrinkage intensit
> plot(x=alphav, y=profilev, t="l",
+   main="Rolling Strategy PnL as Function of Return Shrinkage",
+   xlab="Shrinkage Intensity Alpha", ylab="pnl")
```

Optimal Look-back Interval

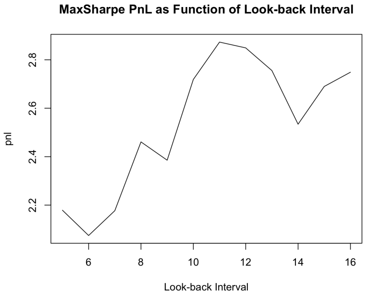
The optimal length of the look-back interval can be determined using *backtesting*.

The optimal length of the look-back interval for this portfolio of stocks is equal to `lookb=10` months, which roughly agrees with the research literature on momentum strategies.

The dependence on the length of the *look-back interval* is an example of the *bias-variance tradeoff*.

If the *look-back interval* is too long, then the data has large *bias* because the distant past may have little relevance to today.

But if the *look-back interval* is too short, then there's not enough data, and estimates will have high *variance*.



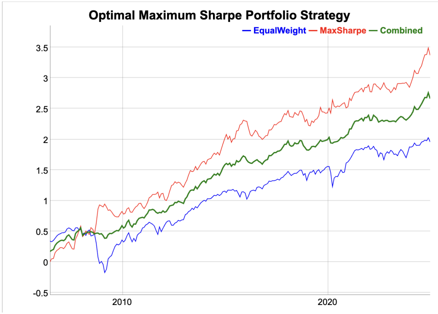

```
> # Create list of model parameters
> controll <- HighFreq::param_portf(method="maxsharpe",
+       dimax=dimax, alpha=alphac)
> # Perform backtest over look-backs
> lookbv <- seq(from=5, to=16, by=1)
> pnls <- mclapply(lookbv, function(lookb) {
+   startp <- c(rep_len(0, lookb), endd[1:(npts-lookb)])
+   startp <- (startp - 1) ; startp[startp < 0] <- 0
+   HighFreq::roll_portf(retx=retx, retp=retp,
+     startp=startp, endd=endd, controll=controll)
+ }, mc.cores=ncores)  ##  end mclapply
> profilev <- sapply(pnls, sum)
> lookb <- lookbv[which.max(profilev)]
```

```
> # Plot of rolling strategy PnL as a function of look-back interval
> plot(x=lookbv, y=profilev, t="l", main="MaxSharpe PnL as Function
+   xlab="Look-back Interval", ylab="pnl")
```

Optimal Portfolio Optimization Strategy

The optimal *portfolio momentum* strategy, with dimension reduction and return shrinkage, has both a higher *Sharpe* ratio and higher absolute returns than the equal weight portfolio.

Combining the optimal portfolio momentum strategy with the equal weight portfolio results in an even higher *Sharpe* ratio.




```
> # Calculate the out-of-sample Sharpe and Sortino ratios
> pnls <- pnls[[whichmax]]
> pnls <- pnls*sd(retew)/sd(pnls)
> wealthv <- cbind(retew, pnls, (pnls + retew)/2)
> colnames(wealthv) <- c("EqualWeight", "MaxSharpe", "Combined")
> sqrt(252)*sapply(wealthv, function(x)
+   c(Sharpe=mean(x)/sd(x), Sortino=mean(x)/sd(x[x<0])))
> # Dygraph the cumulative wealth
> dygraphs::dygraph(cumsum(wealthv)[endw],
+   main="Optimal Maximum Sharpe Portfolio Strategy") %>%
+   dyOptions(colors=c("blue", "red", "green"), strokeWidth=1) %>%
+   dySeries(name="Combined", label="Combined", strokeWidth=2) %>%
+   dyLegend(show="always", width=300)
```

Homework Assignment

Required

Study all the lecture slides in `FRE7241_Lecture_6.pdf`, and run all the code in `FRE7241_Lecture_6.R`

Recommended

- Read about *estimator shrinkage*:
  Aswani Regression Shrinkage Bias Variance Tradeoff.pdf
  Blei Regression Lasso Shrinkage Bias Variance Tradeoff.pdf
- Read about *optimization methods*:
  Bolker Optimization Methods.pdf
  Yollin Optimization.pdf
  DEoptim Introduction.pdf
  Ardia DEoptim Portfolio Optimization.pdf
  Boudt DEoptim Portfolio Optimization.pdf
  Boudt DEoptim Large Portfolio Optimization.pdf
  Mullen Package DEoptim.pdf
- Read about *momentum*:
  Bouchaud Momentum Mean Reversion Equity Returns.pdf